Global-to-Local Spatial–Spectral Awareness Transformer Network for Hyperspectral Anomaly Detection

Xu He, Shilin Zhou, Qiang Ling, Miao Li, Zhaoxu Li, Yuyuan Zhang, and Zaiping Lin

***Abstract*— Hyperspectral anomaly detection (HAD) is one of the momentous technologies in the field of Earth observation and remote sensing monitoring. Profiting from puissant deep feature extraction abilities, deep convolutional networks (DCN) perform excellently in the HAD domain. Nevertheless, limited by the restriction of unique local receptive fields, DCN-based detection methods struggle to catch the long-range dependence from a global perspective. In contrast, vision transformers (ViTs) perform better in global feature extraction but still disregard the local dependence properties. To this end, we proposed a novel method entitled the global-to-local spatial–spectral awareness transformer (G2LSSAT) network, in which the global transformer block (GTB) and local transformer block (LTB) are deployed in sequence to capture deep reconstruction characteristics from the global view to the local view in a spatial–spectral domain. In particular, the GTB is designed to explore the global spatial–spectral characteristics that are dependent on a crossbar-based global sparse attention module. Furthermore, the global glanced image is divided into multiple local patches and the LTB is devised to learn the local spatial–spectral features supported by a patch-based local self-invisible attention module. In addition, considering that the abnormal pixels always be unexpectedly reconstructed with the conventional self-attention module in ViTs, we introduce a invisible diagonal mask (IDM), which is embedded into the LTB module, to overshadow each pixel itself in the receptive field and reconstruct itself based on global and local dependent spatial–spectral features. Extensive experimental results on six datasets illustrate the superiority of the proposed G2LSSAT compared with other state-of-the-art detectors.**

***Index Terms*— Invisible diagonal mask (IDM), global-to-local, hyperspectral anomaly detection (HAD), self-attention, vision transformer (ViT).**

I. Introduction

HYPERSPECTRAL remote sensing imaging technology is one of the cutting-edge technologies in the field of remote sensing for Earth observation. Compared to detection systems that use optical single channels such as visible light, infrared, and ultraviolet, hyperspectral imagers can obtain spectral curves that reflect the "fingerprint" characteristics of targets to effectively identify target types without geometric or textural features [1], [2], [3]. Hyperspectral images (HSIs) have higher spectral resolution and can reach the nanoscale, allowing for the detection and recognition of targets that could not have been effectively detected in multispectral remote sensing images [4], [5], [6]. Among the numerous remote sensing image interpretation tasks based on hyperspectral imaging systems, the hyperspectral object detection task has attracted increasing attention from scholars and researchers owing to its great application value in civil and military fields, such as environmental monitoring [7], [8], resource exploration [9], [10], smart agriculture [11], [12], [13], and military reconnaissance and surveillance [14], [15], [16]. In practical research, the object detection technology for HSIs can be divided into two categories: target detection pattern and anomaly detection (AD) pattern. To be precise, hyperspectral target detection (HTD) mainly compares the spatial–spectral features of each test pixel with the spectral clusters of the background pixels and the prior target spectral clusters. Then, these differences will be mapped into a certain kind of characteristic parameter to reflect the abnormality degree of each test pixel. In contrast, hyperspectral AD (HAD) intends to develop the spectral and spatial distribution differences between the target and the surrounding background to locate the target whose prior spectrum is in an unknown state. HAD task is more in line with practical needs and has greater application prospects because it has freed itself from dependence on the target prior information. Nevertheless, affected by the limitations in imaging conditions such as noise-jamming environments, long-distance imaging, low spatial resolution, and so on, the distinguishability between abnormal targets and heterogeneous backgrounds significantly diminished, which gradually increased the difficulty of background modeling and anomaly abstraction. Thereupon, differentiating anomalies from sophisticated backgrounds is still a hard nut to crack in HSI detection systems. In recent years, more and more scholars have been involved in the research work of HAD, and numerous outstanding algorithms have sprung up like mushrooms. To the best of our knowledge, existing mainstream HAD methods can be divided into four categories,

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i.e., data statistics-based methods, data representation-based methods, data factorization-based methods, and deep learning-based methods.

A. Data Statistics-Based Model

The data statistics-based model was the most classic and time-honored HAD fashion. This kind of method always assumes that the background of HSIs follows an inherent statistical distribution pattern, while the distribution of anomalies significantly deviates from the background distribution, thus screening out anomalies based on this basic principle [17]. Specifically, it will first initialize the statistical model of the assumed background, and then select a certain number of pixels to estimate the parameters of the model. Finally, this model will determine whether the tested pixel is abnormal based on the distance measurement results between the tested pixel and the background distribution. For instance, the most representative method of the data statistics-based model belongs to the Reed–Xiaoli (RX) [18] algorithm, which utilizes the Mahalanobis distance as distance measurement criteria and postulates that the background distribution complies with the multivariate Gaussian distribution. Based on this simple and efficient RX baseline model, many researchers [19], [20], [21], [22] proposed numerous improved and extended algorithms for complicated scenes. Specifically, Kown and Nasrabadi [23] introduced the kernel function and maps the original HSI data into a high-dimensional feature space to enhance the separability of anomalous targets and backgrounds in high-dimensional space. Then, Zhou et al. [24] proposed the cluster kernel RX detector (CKRXD), which effectively reduced the computational complexity of the kernel RX method by replacing each pixel with the cluster centers of the background pixels. Matteoli et al. [25] devised a local adaptive kernel density estimation to model the background distribution and reduce the interference of clusters. Guo et al. [26] proposed the linear filter-RX detector (LFRXD) to filter the abnormal targets and noise for better background covariance matrix estimation and higher detection accuracy. Tu et al. [27], [28] developed the density peak clustering method to eliminate the contamination of anomalies in the process of background modeling. Chang [29], [30], [31] implemented a novel dummy variable trick to transform hyperspectral target detectors into anomaly detectors. Furthermore, Chang et al. [32], [33], [34] developed the versatile effective anomaly space (EAS) scheme to boost the anomaly detectability of anomaly detectors. However, the bulk of data statistics-based models depend heavily on the prior assumptions of data distribution, which restricts their applicability in sophisticated imaging scenarios.

B. Data Representation-Based Model

The data representation-based model will construct the background dictionary and corresponding sparse coefficient matrix of HSIs to determine whether the tested pixel belongs to an abnormal target according to the reconstruction error of the tested pixel without resorting to Gaussian assumptions about the background distribution. Its core mechanism originates from a fundamental theory, that is, each background pixel can be represented by the spatial neighborhood pixels or background dictionaries, but abnormal pixels cannot. In the light of different constraint modes of the representation coefficient, the data representation-based models can be classified as sparse representation (SR)-based models [35], [36] and collaborative representation (CR)-based models [37], [38]. The SR-based model structures the overcomplete dictionary based on the background pixels and estimates the sparse coefficient, which is constrained by the l_1 norm. Then, the abnormal pixels will be screened out due to significant errors in the reconstruction process. Based on the basic SR-based model proposed by Chen et al. [39], many improved HAD algorithms have been designed to boost detection performance. For example, Zhu and Wen [40] improved the performance of the SR-based model by incorporating accurate background estimation and adaptive weighting strategy into the fundamental SR-based HAD framework. Ling et al. [36] applied the nonnegativity and sum-to-one constraints on the sparse coefficient matrix to endow with the practical physical significance of the SR-based method. Moreover, Zhao et al. [41] designed a sparsity score estimation detector to strengthen the discrepancy between the learned anomaly dictionary and the background dictionary. Similar to the SR-based model, the CR-based model considers the spatial–spectral information between pixels and imposes the l_2 norm on the representation coefficient. Based on this fundamental principle, Vafadar and Ghassemian [42] further considered the statistical distribution of background to remove the outliers before the CR operation for pollution-free background dictionary construction. In addition, Hou et al. [43] applied the least squares method to avoid interference from anomalies in the background dictionary construction, and then the spatial saliency analysis was subsequently performed to improve the detection performance. Tan et al. [44] combined the sliding dual-window scheme and CR strategy mechanism to improve the detection accuracy and boost the speed of operation. Chang and Ghamisi [45] devised a novel CR-based model variant, in which the background and anomaly union dictionary are introduced to extract reliable representation samples and remove anomalies or noise contamination. Similarly, Wu et al. [46] presented a relaxed CR-based model that combined the k-nearest neighbor method to construct more a practical nonglobal dictionary for more reliable background representation solutions. Nevertheless, data representation-based methods still easily lose effectiveness in real scenarios due to predetermined parameter selection and tweaking, which reduces the generality of the methods.

C. Data Factorization-Based Model

The data factorization-based model will implement AD by decomposing the hyperspectral cube in different dimensions, which are mainly classified as matrix factorization-based models and tensor factorization-based models. The matrix factorization-based model leverages the theoretical basis that the anomalies in HSIs often present low target-to-image pixel ratios and spatial sparsity, while the background in HSIs is always monotonous and manifested as low-rank attributes. With that in mind, the low-rank and sparse matrix factorization can be exploited in the HAD field. For example,

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Xu et al. [47] represented the background pixels by the low-rank matrix of the background dictionary and excavated the local difference characteristics of anomaly targets by the sparsity constraint. Huyan et al. [48] proposed a potential anomaly and background double dictionary construction method to better discriminate anomalies from background and noise interference. Considering that anomalies and noise have analogous sparse characteristics, the detection results are always mixed with false alarms (FAs) of noise. To this end, Li et al. [49] incorporated the mixture of Gaussian (MoG) method into the matrix factorization-based model, which effectively distinguished between anomalies and noise in a sparse matrix and improved the detection accuracy. Chen et al. [50] presented a simple and effective data decomposition method, called component decomposition analysis (CDA) which better separated anomalies from HSI data. In addition, Chang and Chen [51] also proposed a novel sparsity density peak (SDP) trick to represent the HSI data and extract anomalies in the sparse space. The tensor factorization-based models [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62] take advantage of the 3-D structure of the HSI data cube and achieve high performance in consideration of the 2-D spatial features and 1-D spectral features. Inspired by the mechanism of high-order tensor decomposition, Wang et al. [52] exploited the classical CANDECOMP/PARAFAC (CP) factorization to reconstruct the background based on the principle that the background tensor can be decomposed into the sum of numerous rank-1 tensors. In addition, Zhang and Wen [53] implemented Tucker factorization to achieve the core tensor and corresponding factor tensor, whose principal component can serve as a basis for judging anomalies and backgrounds. Furthermore, Xu et al. [54] introduced tensor singular value decomposition (t-SVD) to resolve the matricization problem of Tucker factorization. However, the tensor nuclear norm (TNN) used in [54] will result in an over-shrinkage problem and deteriorate the detection performance. To solve this issue, He et al. [55] introduced a novel ϵ-shrinkage TNN to obtain a more compact convex closure of the tensor tubal rank. Xiao et al. [63] introduced the optimal clustering framework-based enhanced tensor low-rank (OCF-ETLR) factorization model to select effective bands and reduce the interference from redundant bands. Besides that, Feng et al. [56] employed the tensor ring factorization and total variation constraint to extract the low-rank and piecewise smoothness attributes of the background in the spatial and spectral domains. Xiao et al. [64] also applied the total variation constraint and explored a multiscale entanglement renormalization ansatz (MERA) based tensor factorization algorithm to decompose the HSI tensor into anomalies and background parts. Notwithstanding, these methods only exploit handcrafted shallow features and image attributes such as low-rank and sparsity characteristics, lacking significant deep context features devoted to discriminate anomalies from the background.

D. Deep Learning (DL)-Based Model

Unlike the above three methods, DL-based models are prior knowledge agnostic and do not depend upon data distribution assumptions. Actually, off-the-shelf DL-based methods mostly adopt the reconstruction-based self-supervised data fitting pattern to precisely reconstruct the background and suppress anomalies with large reconstruction errors [65], [66]. To the best of our knowledge, the mainstream DL-based HAD solutions include convolution neural network (CNN)-based models and generative adversarial network (GAN)-based models. The CNN-based model exploits the formidable feature extraction capability of CNNs to extract the spatial–spectral features and reach effective reconstruction. As the inventors of CNN-based models, Li et al. [67] trained the network with differential pixel pairs and similar pixel pairs samples to determine whether the test sample is abnormal. Moreover, Fu et al. [68] solved the HAD problem with a plug-and-play denoising network model and employed CNN regularization to eliminate background noise. Wang et al. [69] applied transfer learning to obtain more training samples and effectively address the prior knowledge agnostic problem. In addition, the autonomous AD (Auto-AD) [70] implemented a fully convolutional autoencoder network with a skip connection structure to obtain a reconstruction error map with high contrast between anomalies and the background. Based on the autoencoder network framework, Liu et al. [71] designed a dual-frequency autoencoder model which converted the hyperspectral data into low- and high-frequency constituents and parallelly extracted anomalies from these two constituents. Meanwhile, Jiang et al. [72] developed an autoencoding density estimation network which estimated the energy density of anomalies with an end-to-end unsupervised learning manner. Li et al. [73] employed an interpretable model-driven deep mixture network which combined the autoencoder-based low-dimensional manifold background learner and convex relaxation optimization with anomaly sparse constraints. The GAN-based methods effectively model the background distribution and reconstruct the background by means of the powerful data distribution learning ability of GAN. For example, Jiang et al. [74] proposed an unsupervised discriminative reconstruction constrained GAN (DRCGAN), which employed an autoencoder to realize the discriminative reconstruction constraint for GAN and accurate background pixel distribution learning. Based on the variational background inference strategy, Wang et al. [75] achieved excellent background modeling and spectrum reconstruction effect in light of the GAN-based network framework. Furthermore, Jiang et al. [76] devised a spectral-constrained GAN, called WeaklyAD, for the reconstruction task of HSIs, where the anomaly and the background can be better discriminated with a higher reconstruction error of the anomaly target. Similarly, Li et al. [77] presented a novel sparse HAD network that integrated the sparse coding strategy into the GAN-based reconstruction network and improved the robustness of this weakly supervised HAD network. Jiang et al. [78] implemented an unsupervised, low-rank embedded network (LREN), where the latent layer cooperated with the Gaussian mixture model.

E. Motivation and Contributions

Through an in-depth analysis of the above DL-based methods, it can be perceived that the pivotal proposition of

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the DL-based models rests on the construction of differential characteristics representation between backgrounds and anomalies. Therefore, the employment of reconstruction error, minimization can go along with this central foothold and achieve exceptional detection performance in certain scenarios based on the basic assumption, where the background part can be correctly reconstructed by the network while anomaly targets cannot [79]. However, due to inevitable abnormal sample contamination during network training, off-the-shelf detection network mostly has the flaw of inevitable memory storage for anomalies [80]. Therefore, it is not always pledged that anomalies cause emphatically higher reconstruction errors than the background in the existing DL-based methods [81], [82]. To this end, specific feature guiding constraints should be applied for the background and anomalies to enhance the background reconstruction while weakening the anomalies reconstruction. In recent, several relevant DL-based literature about imposing specific constraints on HSIs guiding reconstruction have emerged. For example, Auto-AD [70] exploited a novel adaptive weighting loss that dynamically utilized the reconstruction errors of anomalies to determine the adaptive weighting loss for better anomalies suppression during network training. Moreover, inspired by representation-based methods, the deformable convolutional autoencoder (DCAE) [83] designed a background guidance module, which restrained the anomalies reconstruction by guiding the parameter updating of the decoder in light of the elaborate background dictionary representation. However, the deployment of the above constraints only a trifle retards the issue of anomalies reconstruction problem due to the neglect of self-supervised memory learning for anomalies during network training. In contrast, exploring the anomalies overshadowing strategy can resolve the anomalies reconstruction issue during self-supervised learning. Specifically, Wang et al. [84], [85], [86] proposed the invisible-spot self-supervised learning network (BS^{3}LNet), which generated a high-quality reconstruction error map owing to the employment of anomalies overshadowing-based invisible-spot architecture. Nevertheless, restricted by the intrinsic local receptive fields of CNN framework, almost all existing DL-based AD methods, including BS^{3}LNet, were flawed in capturing the long-range dependence and nonlocal content features from a global perspective. In conclusion, the limitations of existing DL-based AD methods can be summarized as the inevitable anomalies reconstruction issue and intrinsic CNN local receptive fields restriction. To this end, the research in this article offers a new perspective on leading transformers into HAD tasks. Specifically, as shown in Fig. 1, the target discovery process in the human vision system (HVS) actually adheres to the global-to-local attention pattern which combines global glance attention with local refining attention. Meanwhile, as expressed in Fig. 2, both local spatial–spectral differences and nonlocal similarity features are momentous clues for reconstruction learning in the HAD task. Therefore, inspired by this HVS attention mechanism, we propose a global-to-local spatial–spectral awareness transformer (G2LSSAT) network paradigm for the HAD task. First, by means of the exceptional long-range feature extraction capability of the

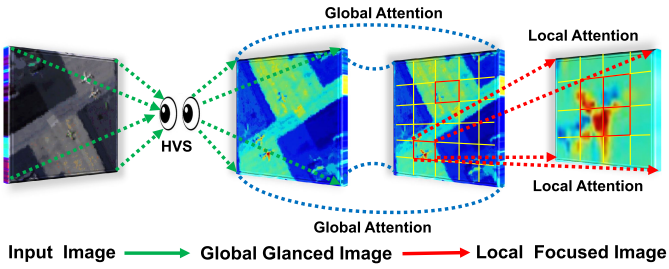
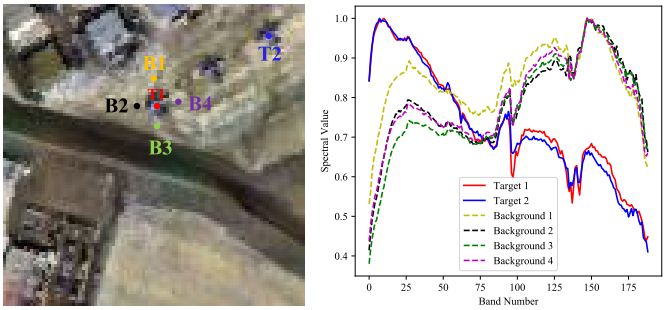


Fig. 1. Schematic of the visual attention mechanism of the HVS. This visual attention mechanism mainly adopts a global-to-local pattern. After observing an image, HVS will first scan the entire image through a global attention mechanism to obtain the global glanced image and key local regions of interest. Then, HVS will further refine the feature extraction and attention of the key local regions in a local focused image.



(a) (b)

Fig. 2. Examples of the local and nonlocal spatial–spectral relationships among pixels in HSIs. T1 and T2 represent the nonlocal relevant pixels with spatial–spectral similarity. B1, B2, B3, and B4 are the pixels surrounding the local neighborhood of T1. (a) and (b) Illustrate the spatial positional relationships and their respective spectral curves of corresponding pixels. The spectral curve of T1 is easily discriminated from its adjacent pixels and determined as abnormal. Meanwhile, since T1 and T2 present extremely similar spectral radiation and spatial morphological characteristics, long-range pixel T2 can also be judged as an outlier if T1 belongs to the anomaly part.

transformer, we first implement a global transformer block (GTB) that extracts the global glanced spatial–spectral features based on a crossbar-based global sparse self-attention module. Then, similar to the global-to-local object detection mode of HVS, the global glanced image is divided into multiple local patches, followed by the employment of a local transformer block (LTB) to abstract the local spatial–spectral features supported by a patch-based local self-invisible attention module. Moreover, a novel anomalies overshadowing self-supervised learning strategy is devised to alleviate the anomalies reconstruction problem. Specifically, we incorporate an elaborate invisible diagonal mask (IDM) into the self-attention module, which overshadows each abnormal pixel or background pixel and implements reconstruction learning based on the dependent spatial–spectral features from the global and local receptive fields. In our experiments, we conduct extensive contrastive experiments and ablation studies on multiple datasets to verify the effectiveness of the proposed G2LSSAT. To summarize, the major contributions of our article are concluded as follows.

1) We develop a brand-new global-to-local HAD framework based on the choreographed GTB-to-LTB transformer network structure, which effectively extracts deep

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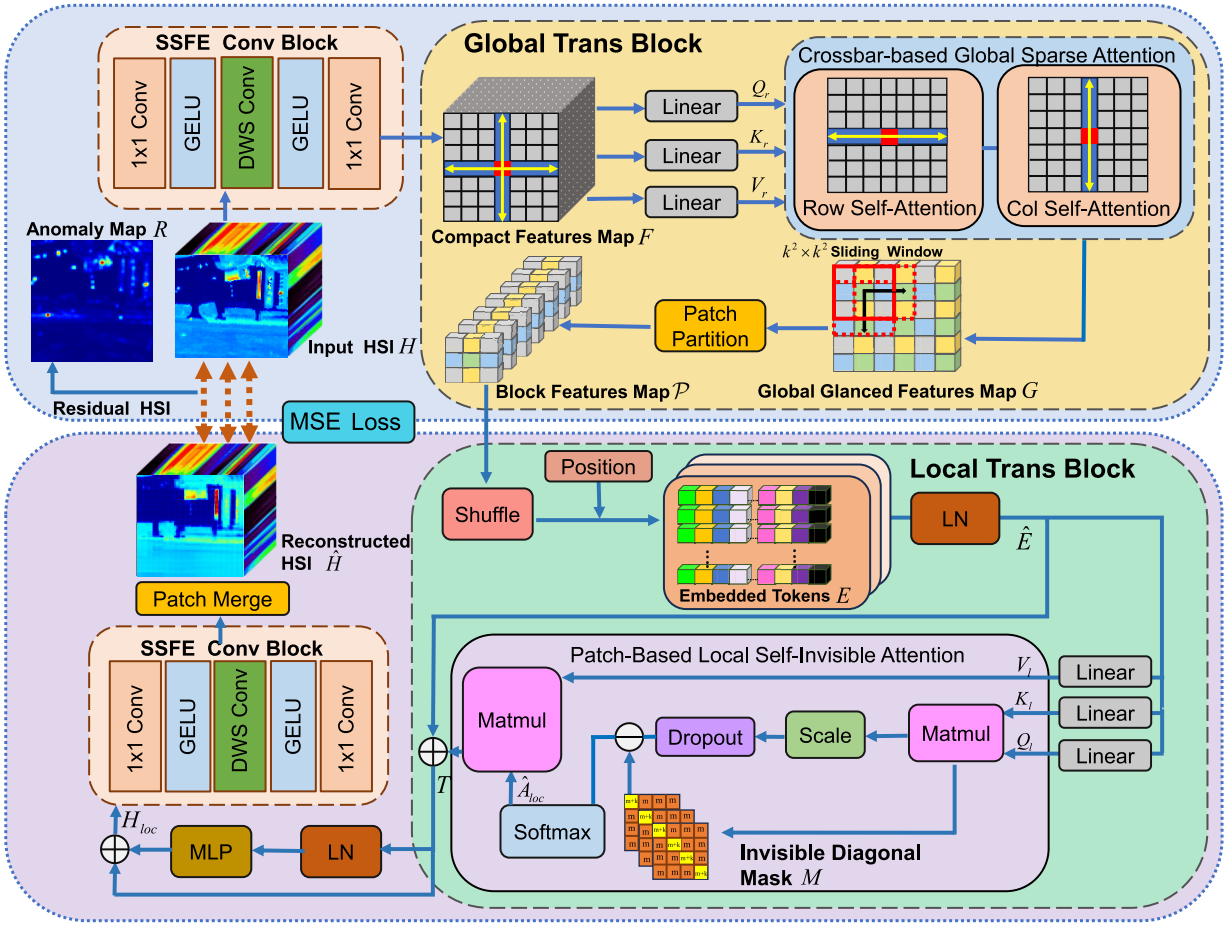


Fig. 3. Overall network framework of G2LSSAT. The proposed method is devised based on a global-to-local transformer architecture. Conv and Trans are abbreviations for convolution and transformer. Linear, LN, and MLP represent the LPL, LayerNorm layer, and multilayer perceptron, respectively.

spatial–spectral reconstruction features from a global perspective to a local perspective.

2) We introduce the crossbar-based global sparse attention module and patch-based local self-invisible attention module and embed them into the GTB and LTB for nonlocal and local spatial–spectral characteristics learning.
3) We develop novel anomalies overshadowing-based self-supervised network learning strategy to address the anomalies reconstruction issue based on a IDM.
4) We conduct copious comprehensive experiments on six hyperspectral datasets to test and verify the superiority of the proposed G2LSSAT for HAD.

The reminder of our article is organized as follows. First, we introduce the related work and motivation of our work in Section I. In Section II, we present the details of the proposed G2LSSAT. In Section III, the experimental results and analysis are provided. Finally, the conclusion is drawn in Section IV.

II. Proposed Methodology

Taking inspiration from the global-to-local vision attention mechanism of HVS and the transcendent capability of the transformer in extracting global dependent features, we present a novel G2LSSAT. In this section, we illustrate the proposed G2LSSAT and explain the implementation details.

A. Overall Network Architecture

The overall network structure diagram of the proposed G2LSSAT is shown in Fig 3. Let the input HSI data cube be represented as $H \in \mathbb{R}^{b\times h\times w}$, where h and w represent the height and width of the input HSI, and b is the number of spectral bands. For the input HSI, we first employ a spatial–spectral features enhancement (SSFE) convolution block to enhance the spatial and spectral features representation. Specifically, the SSFE convolution block consists of 1×1 convolution layers, Gaussian error linear units (GELUs) layers, and depth-wise separable (DWS) convolution layers [87]. In addition, the DWS convolution layer includes a 3×3 convolution layer (depth-wise convolution) and a 1×1 convolution layer (point-wise convolution). Actually, the DWS convolution layers can effectively strengthen spatial–spectral features from spatial–spectral dimensions and increase the generalization and trainability of the model. Therefore, after the input HSI H is fed into the SSFE convolution block, we can obtain the compact feature map $F \in \mathbb{R}^{b\times h\times w}$ as follows:

$$F = \mathcal{C}_{1\times1}(\mathcal{G}(\mathcal{DWS}(\mathcal{G}(\mathcal{C}_{1\times1}(H))))) \tag{1}$$

where $\mathcal{C}_{1\times1}$, $\mathcal{G}$, and $\mathcal{DWS}$ represent the 1×1 convolution layer, GELU layer, and DWS convolution layer, respectively. Subsequently, the compact feature map F is fed into the

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Algorithm 1 Training Procedure of the G2LSSAT

Input: (1) the input HSI data $H \in \mathbb{R}^{b\times h\times w}$;
(2) maximum iteration T_{max};
(3) batch size n;
(4) sliding windows stride k.

Procedure:
Initialize the network with random weights.
for epoch in [1,2,...,T_{max}] **do**
  Strengthen spatial-spectral features according to 1.
  Update non-local spatial-spectral features based on global transformer block.
  Construct the local blocks according to the sliding windows-based patch partition in 2.
  Shuffle the local blocks and select the batch of shuffled blocks as training samples.
  for each sample batch n **do**
    Update local spatial-spectral characteristics based on local transformer block.
    Strengthen spatial-spectral features based on 1.
    Train the network by minimizing loss in 3.
    Update network parameters with ADAM optimizer.
  end
  Merge patches according to the position calibration matrix of the shuffled blocks.
end
Output: Network parameters.

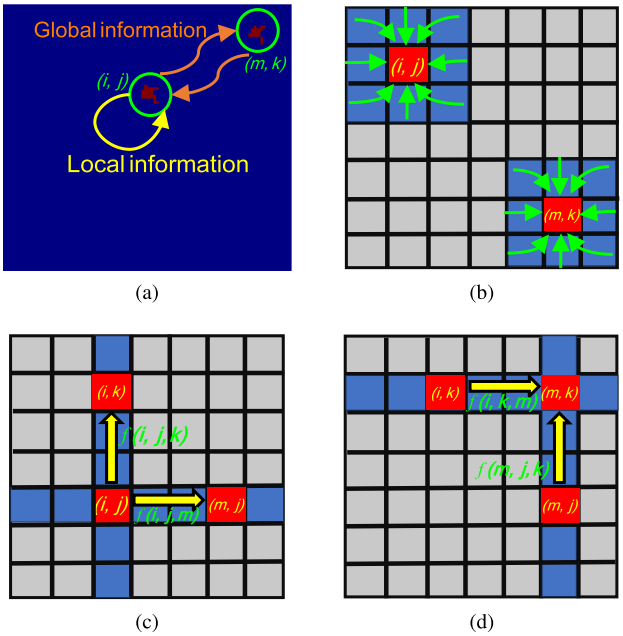


Fig. 4. Diagrammatic sketch of the global-to-local mechanism in our G2LSSAT. (a) Global and local information dependent relationship. The local neighborhood pixels surrounding the central pixel and nonlocal related pixels can be utilized for reconstruction. (b) Local neighborhood pixels reconstruction mechanism. The blue pixels denote the neighborhood pixels which can be used to reconstruct the central pixel. (c) and (d) Represent the relating pattern between spectral vector located in spatial coordinate (i, j) and spatial coordinate (m, k). It can be seen that the related spectral vector located in spatial coordinate (i, k) and spatial coordinate (m, j) can be captured by the proposed crossbar-based global sparse attention. The pertinence relation between (i, j) and (m, k) can be indirectly represented by the aid of (i, k) and (m, j). Meanwhile, it can be easily deduced that the nonlocal information cannot be obtained by single row or column self-attention.

GTB to extract the global features and obtain the glanced feature map G based on a pivotal crossbar-based global sparse attention module. Subsequently, we implement a $k^2\times k^2$ sliding window to partition G into multiple sliding HSI local blocks $\mathcal{P} = \{p_i\}_{i=1}^{N}$, with the sliding stride being k. In particular, the spatial size of each local block is $k^2 \times k^2$ and the total number of local blocks n_b is formulated as

$$n_b = n_w \times n_h = \prod_{i\in(0,1)} \left\lfloor \frac{s_i + 2p_b - d_b(k^2-1) - 1}{k} + 1 \right\rfloor \tag{2}$$

where s_i $(i = 1, 2)$ represents the spatial size of the given HSI, i.e., the width and height of the HSI. Notably, p_b and d_b denote the padding and dilation parameters which are set to 0 and 1 in the unfold operation. Therefore, we can obtain the local block feature map with a size of $\mathcal{P} \in \mathbb{R}^{n\times k^2\times k^2\times b}$, where n denotes the number of batch size. Then, the block feature map $\mathcal{P}$ will be shuffled along the first dimension to enhance robustness during network training. In addition, the block feature map will be divided into multiple local patches and reshaped as an embedded token with the size of $E \in \mathbb{R}^{n\times(k^2\times k^2)\times b}$. The position calibration matrix $C \in \mathbb{R}^{n_w\times n_h}$ will storages the position information of the shuffled block feature map for the following patch merge process. Then, the embedded token will be normalized by the LayerNorm operation and fed into the local transformer block to capture the local spatial–spectral reconstruction features supported by a patch-based local self-invisible attention module. Finally, the updated block feature map will be further enhanced by employing SSFE operation. For the loss optimization of the network, our G2LSSAT network is trained using the mean squared error (mse) loss function according to the practical experimental results in Section III-G

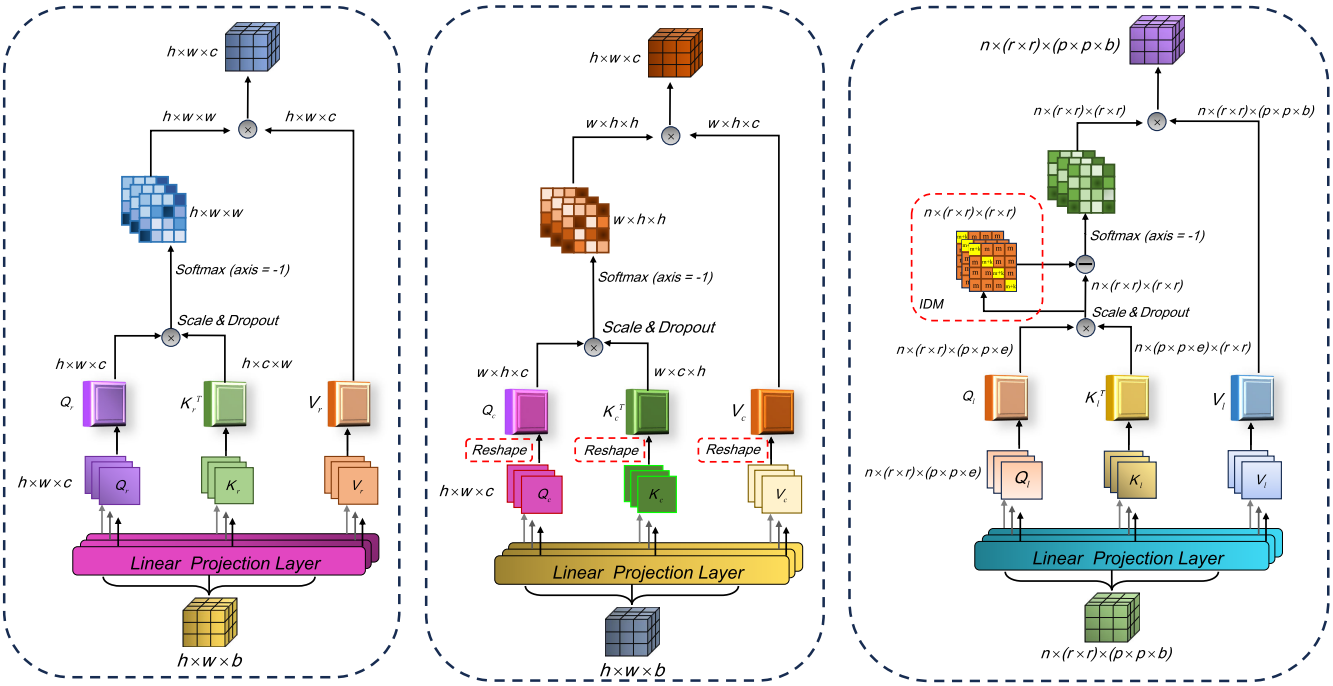
$$\begin{aligned}\hat{\psi} &= \arg\min_{\psi} \mathscr{L}(\psi) \\ &= \arg\min_{\psi} \|\mathscr{F}(H;\psi) - H\|_2^2\end{aligned} \tag{3}$$

where $\mathscr{F}(H;\psi)$ represents the overall network optimization function and ψ refers to the learnable parameters in our G2LSSAT. In the testing phase, the HAD result will be given according to the reconstruction residual R between the input HSI H and reconstructed HSI $\hat{H}$ as follows:

$$R_{i,j} = \left\|\mathbf{h}_{i,j} - \hat{\mathbf{h}}_{i,j}\right\|_2 \tag{4}$$

where $\mathbf{h}_{i,j} \in \mathbb{R}^{b\times 1}$ and $\hat{\mathbf{h}}_{i,j} \in \mathbb{R}^{b\times 1}$ represent the pixels at location (i, j) of H and $\hat{H}$, respectively. Specifically, $R_{i,j}$ denotes the anomaly scores of the corresponding pixel located at (i, j) of the final AD map $R \in \mathbb{R}^{h\times w}$. Note that all patches can be exploited to calculate the reconstruction error loss and generate the final AD map in the training and testing stage. The critical processes of the proposed G2LSSAT are described in Algorithm 1. Meanwhile, we record the feature dimensions of the input and output of each module in Table I.

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(a) (b) (c)

Fig. 5. Detailed network architecture of the crossbar-based global sparse self-attention and patch-based local self-invisible attention module. (a) Row sparse self-attention module. (b) Column sparse self-attention module. (c) Patch-based local self-invisible attention module.

B. Global Transformer Block

As shown in Fig. 4(a), we can utilize the nonlocal content similarity features, i.e., two planes with similar features at a long distance, to improve the ability of the network over anomalies judgment. Therefore, as presented in Fig 3, for the enhanced compact feature map F, we first fed it into a GTB to explore the nonlocal dependent spatial–spectral features. Specifically, the core composition of the GTB focuses on a crossbar-based global sparse attention module. In a conventional transformer, the multihead self-attention layers serve as the prominent network architectural components that can extract long-range context information. Nevertheless, multihead self-attention is of quadratic complexity $O(N^2)$ [88], which increases computational burden and training difficulty. To tackle this issue, we develop a novel crossbar-based global sparse attention, which is composed of horizontal row self-attention and vertical column self-attention module. The detailed network diagrammatic sketch of the proposed crossbar-based global sparse attention is depicted in Fig. 5(a) and (b). Specifically, for the horizontal row self-attention, the given HSI data cube $F \in \mathbb{R}^{h\times w\times b}$ will be projected into query $Q_r \in \mathbb{R}^{h\times w\times c}$, key $K_r \in \mathbb{R}^{h\times w\times c}$, and value $V_r \in \mathbb{R}^{h\times w\times c}$ by linear projection layers (LPLs)

$$Q_r = FW_{Q_r}, \quad K_r = FW_{K_r}, \quad V_r = FW_{V_r} \tag{5}$$

where $c = b/2$ in our method. $W_{Q_r} \in \mathbb{R}^{b\times c}$, $W_{K_r} \in \mathbb{R}^{b\times c}$, and $W_{V_r} \in \mathbb{R}^{b\times c}$ represent the learnable weight matrix. Subsequently, the projected query Q_r and key K_r are rearranged to produce the following row attention map A_{row} by the row self-attention mechanism:

$$A_{\text{row}} = \text{Softmax}\left(\frac{Q_r K_r^T}{\sqrt{c}}\right) \in \mathbb{R}^{h\times w\times w} \tag{6}$$

where Softmax(·) denotes the softmax function along the last axis of the data cube, $\sqrt{c}$ represents the scale factor. Intuitively, as depicted in Fig. 6, we present the sparse row spatial–spectral self-attention mechanism in detail. Specifically, for the spectral vectors located at spatial coordinates (i, j) in projected query Q_r, all spectral vectors located on the same row, i.e., the same row matrix $Q_i \in \mathbb{R}^{w\times c}$, will be included in the self-attention calculation. The same row Q_i matrix will be multiplied by the transpose matrix K_i^T of the same row matrix K_i in the projected key K_r to generate the ith row self-attention map $A_i \in \mathbb{R}^{w\times w}$. By traversing all rows, the final row self-attention map is given in (6). Meanwhile, the projected value V_r will be weighted by multiplying V_r and the row self-attention map, which models the spatial–spectral correlation between the spectral vector in spatial coordinates (i, j) and all spectral vectors in the same row. It can be noticed from Fig. 5(a) and (b) that the row self-attention map is updated by the dropout operation as $\hat{A}_{\text{row}} = \mathcal{D}(A_{\text{row}})$, where $\mathcal{D}$ denotes the dropout operation. Then, the final output of the row self-attention module can be expressed as $\hat{A}_{\text{row}}V$. Except for the row self-attention, the column self-attention can be obtained by traversing all columns according to a similar principle. Note that the row and column self-attention must be iteratively spliced in series for nonlocal spatial–spectral feature extraction and generating global glanced features map G as indicated in Fig. 4(c) and (d). In other words, a single row or column self-attention cannot capture long-range features. In light of the proposed

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TABLE I
FEATURE DIMENSIONS OF THE INPUT AND OUTPUT OF EACH MODULE

Modules	Operation	Input size	Output size
Spatial-spectral feature enhancement for input HSI	1×1 Convolution	191× 100× 100	191× 100× 100
	GELU	191× 100× 100	191× 100× 100
	DWS Convolution	191× 100× 100	256× 100× 100
	GELU	256× 100× 100	256× 100× 100
	1×1 Convolution	256× 100× 100	191× 100× 100
Global Transformer Block	Row sparse self-attention	191× 100× 100	191× 100× 100
	Linear Projection	191× 100× 100	191× 100× 100
	Column sparse self-attention	191× 100× 100	191× 100× 100
	Linear Projection	191× 100× 100	191× 100× 100
	Patch Partition	191× 100× 100	1156×191× 9× 9
Local Transformer Block	LayerNorm	256× 81× 191	256× 81× 191
	Local self-invisible attention	256× 9× 9× 191	256× 9× 9× 191
	LayerNorm	256× 81× 191	256× 81× 191
	Multi-layer perceptron	256× 81× 191	256× 81× 191
Spatial-spectral feature enhancement for updated HSI	1×1 Convolution	256×191× 9× 9	256×191× 9× 9
	GELU	256×191× 9× 9	256×191× 9× 9
	DWS Convolution	256×191× 9× 9	256×256× 9× 9
	GELU	256×256× 9× 9	256×256× 9× 9
	1×1 Convolution	256×256× 9× 9	256×191× 9× 9
	Patch Merge	1156× 191× 9× 9	191× 100× 100

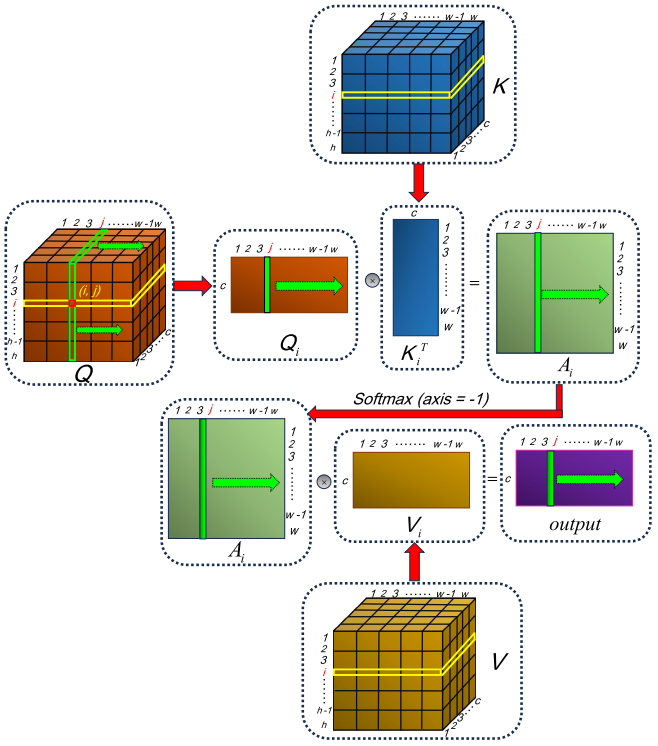


Fig. 6. Sketch map of the calculation process of the row self-attention. The yellow area represents the row matrix O_i where the spectral vector at spatial coordinate (i, j) is located. The green arrow represents the traversal direction of each column of O_i in the process of multiplying matrix O_i by the transpose of K_i.

crossbar-based global sparse self-attention, the computational complexity of global self-attention has been reduced from $O(N^2)$ to $O(N)$ due to the serial iterative alternating implementation of row and column self-attention with complexity $O(N)$ [88].

C. Local Transformer Block

As described in Section II-A, we employ a sliding window to divide the global glanced feature map G into multiple local patches and introduce an LTB to extract local spatial–spectral deep features. In our method, the sliding stride p and window size w are set to k and k^2, where k is a hyperparameter. According to the sliding stride and window size of the sliding window, we can obtain the number of patches $r = w/p$ in each local block. Subsequently, as shown in Fig. 3, the embedded tokens $E \in \mathbb{R}^{n\times(w\times w)\times b}$ will be fed into the LayerNorm and reshaped as $\hat{E} \in \mathbb{R}^{n\times(r\times r)\times(p\times p\times b)}$, where each patch is straightened into a 1-D spectral vector with the size of $(p\times p\times b)$. Then, as shown in Fig. 5(c), the updated embedded tokens $\hat{E}$ will be imputed into the LPL to obtain the local patch query $Q_l \in \mathbb{R}^{n\times(r\times r)\times(p\times p\times e)}$, key $K_l \in \mathbb{R}^{n\times(r\times r)\times(p\times p\times e)}$, and value $V_l \in \mathbb{R}^{n\times(r\times r)\times(p\times p\times b)}$

$$Q_l = \hat{E}W_{Q_l}, \quad K_l = \hat{E}W_{K_l}, \quad V_l = \hat{E}W_{V_l} \tag{7}$$

where $e = b/2$ in our method. $W_{Q_l} \in \mathbb{R}^{b\times e}$, $W_{K_l} \in \mathbb{R}^{b\times e}$, and $W_{V_l} \in \mathbb{R}^{b\times b}$ represent the learnable weight matrix in LPLs. Correspondingly, the local patch-based attention map A can be obtained by multiplying the projected local patch query Q_l by the transpose of key K_l

$$A_{\text{loc}} = \text{Softmax}\left(\frac{Q_l K_l^T}{\sqrt{p\times p\times e}}\right) \in \mathbb{R}^{n\times(r\times r)\times(r\times r)} \tag{8}$$

where $(p \times p \times e)^{1/2}$ represents the corresponding scale factor. Similar to the updating fashion of row and column self-attention module, we incorporate dropout and invisible diagonal masking operations to generate the final patch-based local self-invisible attention map $\hat{A}_{\text{loc}}$. The detailed mechanism of the IDM can be found in Section II-D. Finally, we can

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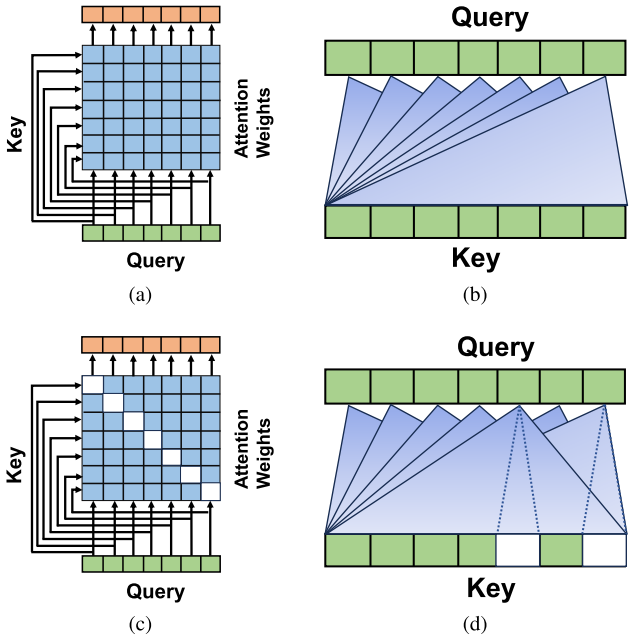


Fig. 7. Diagrammatic sketch of self-attention and the proposed invisible diagonal masked self-attention in our G2LSSAT. The white regions in attention weights represent zero-value regions. (a) Self-attention mechanism. (b) Receptive fields of self-attention. (c) Invisible diagonal masked self-attention mechanism. (d) Receptive fields of invisible diagonal masked self-attention.

obtain the local-focused HSI data cube H_{loc} as follows:

$$\begin{aligned} H_{\text{loc}} &= T + \text{MLP}(\text{LN}(T)) \\ T &= \text{Reshape}(\hat{A}_{\text{loc}} V_l) + \hat{E} \end{aligned} \tag{9}$$

where MLP and LN represent the multilayer perceptron layer and LayerNorm layer, respectively. Moreover, reshape denotes the operation of adjusting $\hat{A}_{\text{loc}} V_l \in \mathbb{R}^{n\times(r\times r)\times(p\times p\times b)}$ as the size of $\text{Reshape}(\hat{A}_{\text{loc}} V_l) \in \mathbb{R}^{n\times(r\times p)\times(r\times p)\times b}$. Then, the SSFE module will be embedded after the local focused feature map to strengthen the spatial–spectral reconstruction representation. Finally, the local focused HSI data cube will be merged to form the output reconstructed HSI with the size of $\hat{H} \in \mathbb{R}^{w\times h\times b}$ according to the position calibration matrix.

D. Invisible Diagonal Mask

In the architecture of the patch-based local self-invisible attention module, we incorporate a IDM to suppress excessive anomalies reconstruction. This section explains the detailed mechanism of the IDM. Specifically, as illustrated in Fig. 7(a) and (b), for conventional self-attention, the attention weight A can be denoted as $\text{Softmax}(QK^T/\sqrt{d}) \in \mathbb{R}^{m\times m}$ based on the linear projected query $Q \in \mathbb{R}^{m\times d}$ and the linear projected value $K \in \mathbb{R}^{m\times d}$. In a nutshell, this type of self-attention mechanism will learn the dependency relationship of each query on all other keys and build the global receptive field. In other words, each feature information is a combination of all other features within the group. In the field of imaging, the self-attention mechanism can enable each patch to contain not only its own subject information but also its relationship information with other patches. Nevertheless, the overall optimization purpose of the reconstruction network is to reconstruct the background and suppress the anomalies to the maximum extent. Therefore, we hope that the network will overshadow each pixel itself in the receptive field and reconstruct itself by means of its relationship information with other pixels. In that case, the anomaly pixel that distinctly differs from the adjacent pixels cannot be dependably restored and generates a larger reconstruction error during network training. From this point of view, it is considered that the diagonal element which describes the data autocorrelation property in the attention weight matrix illustrates the receptive field of each pixel for itself during the network training stage. Therefore, as shown in Fig. 7(c) and (d), the diagonal element in the attention weight matrix A, which measures the degree of self-correlation must be set to 0. To this end, for the attention weight matrix A, we design a novel IDM, which is embedded before the Softmax operation. Let $S = (QK^T)/\sqrt{d} \in \mathbb{R}^{m\times m}$ be represented as follows:

$$S = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1m} \\ s_{21} & s_{22} & \cdots & s_{2m} \\ \vdots & \vdots & & \vdots \\ s_{m1} & s_{m2} & \cdots & s_{mm} \end{bmatrix}. \tag{10}$$

Let m_i denote the maximum value of ith row, the IDM can be represented as follows:

$$M = \begin{bmatrix} m_0 + \infty & m_0 & \cdots & m_0 \\ m_1 & m_1 + \infty & \cdots & m_1 \\ \vdots & \vdots & \ddots & \vdots \\ m_m & m_m & \cdots & m_m + \infty \end{bmatrix} \tag{11}$$

where ∞ represents a large positive value which is set to 300 in our method.

Based on the IDM, the final attention weight matrix can be described as $\hat{A} = \text{Softmax}(S - M)$. On the one hand, we subtract the maximum value of each row from each element in the attention weight matrix to prevent a data overflow issue [89]. On the other hand, we subtract a large value from each value on the diagonal and utilize the Softmax operation, i.e., $\lim_{x\to-\infty} \exp^{-x} = 0$, to zero the diagonal elements.

III. Experimental Results and Analyses

In this section, numerous experiments are implemented to test and verify the effectiveness of our G2LSSAT for the HAD task on six real HSI datasets. Note that all the comparison experiments were conducted on a workstation with Windows 10, Intel[1] Core[2] i9-10980XE CPU@3.00 GHz, 16-GB RAM, and an NVIDIA 3090 GPU with 24-GB memory.

A. Datasets Introduction

In the experiments, we compare our method with nine well-developed detectors on six hyperspectral datasets. The detailed introduction of the six datasets is presented in Table II.

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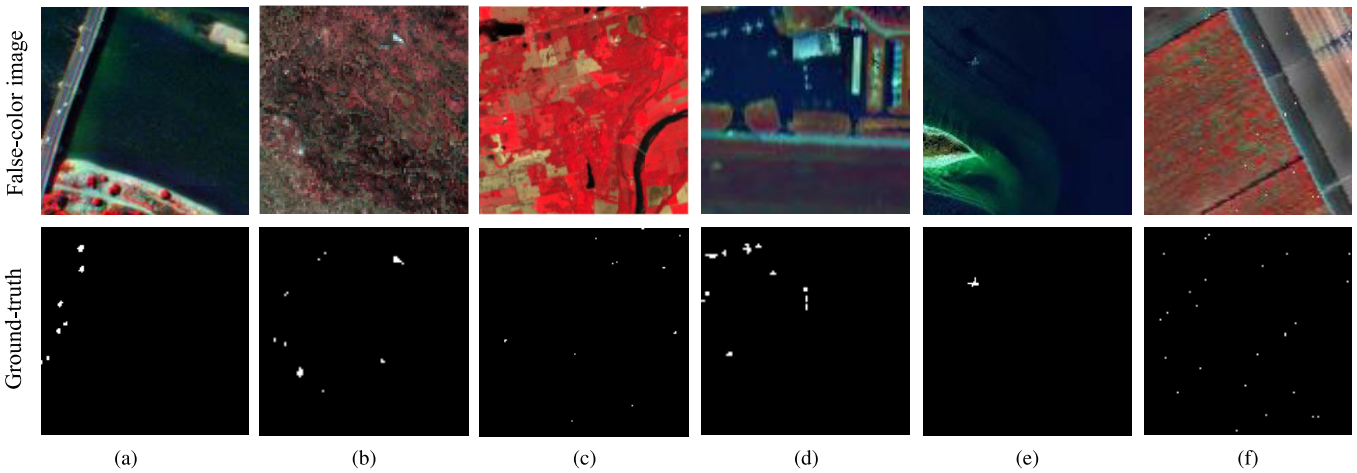


Fig. 8. HSI datasets used in our experiments. The first row and second row represent the pseudo-color image and ground truth of the corresponding dataset. (a) Pavia. (b) Okavango Delta. (c) Indiana. (d) Gainesville. (e) Cat Island. (f) Salians.

TABLE II
INTRODUCTION OF SIX HSI DATASETS. RES. REPRESENTS THE ABBREVIATION OF RESOLUTION

Data Sets	Sensor type	Target Type	Background Type	Spatial size	Target size	Band number/range	Spatial Res.	Abnormal proportion
Pavia	ROSIS-03	vehicles	Beach area	150×150 pixels	68 pixels	102/430-680nm	1.3m	0.30%
Okavango Delta	Hyperion	unknown	Land cover area	100×100 pixels	32 pixels	145/400-2500nm	30m	0.32%
Indiana	Hyperion	storage silos/roofs	Agricultural region	150×150 pixels	17 pixels	149/357-2576nm	10m	0.07%
Gainesville	AVIRIS	ships	Urban area	100×100 pixels	52 pixels	191/400-2500nm	3.5m	0.52%
Cat Island	AVIRIS	ships	Beach area	150×150 pixels	19 pixels	188/400-2500nm	17.2m	0.08%
Salians	AVIRIS	synthetic targets	Agricultural region	120×120 pixels	25 pixels	204/400-2500nm	3.7m	0.17%

1) Pavia Dataset: The first dataset was captured by the Reflective Optics System Imaging Spectrometer-03 (ROSIS-03) sensor over the Pavia city of Italy [90]. The spectral wavelength range of the Pavia dataset involves 188 bands in the range from 430 to 680 nm. Meanwhile, the spatial resolution and spatial size of this dataset come to 1.3 m/pixel and 150 × 150 pixels. As presented in Fig. 8(a), this dataset is composed of the beach background and anomalous vehicle targets with 68 pixels. The total abnormal target pixels account for 0.3% of the entire hyperspectral image.

2) Okavango Delta Dataset: The second dataset is an HSI with a land scene collected by the Hyperion sensor [91] carried by EO-1. It was captured over the area of the Okavango Delta. The wavelengths of the bands in the Okavango Delta dataset range from 400 to 2500 nm. The background area in this dataset consists of observations from 14 different types of land cover. In addition, this dataset is composed of 100 × 100 pixels with 30 m/pixel spatial resolution and 145 spectral bands are preserved to hold discriminative information for each pixel. As shown in Fig. 8(b), 32 anomalous pixels are to be detected in the land background, making up 0.32% of the total amount of pixels in the dataset.

3) Indiana Dataset: The third dataset was collected by the Hyperion hyperspectral sensor over the zone of the State of Indiana, USA [92]. It contains 149 spectral bands in the wavelength range of 357∼2576 nm. Under a 10 m/pixel spatial resolution, this dataset has a spatial size of 150 × 150 pixels. As presented in Fig. 8(c), the 17 abnormal pixels mainly consist of the storage silos and the roofs embedded in the agricultural background region.

4) Gainesville Dataset: The fourth dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the region of Gainesville, Florida, USA [90]. It has 191 spectral bands used for the experiments after erasing the low signal-to-cluster ratio (SCR) and water absorption bands from the incipient dataset. Moreover, the Gainesville dataset is comprised of 100 × 100 pixels, with a spatial resolution of 3.5 m/pixel. As presented in Fig. 8(d), this dataset includes 52 anomaly pixels with the ship attribute to be detected, which accounts for 0.52% of all the pixels.

5) Cat Island Dataset: The fifth dataset with a beach background was collected by the AVIRIS sensor [90]. As described in Table II, the spatial size and number of spectral bands are 150 × 150 pixels and 188, respectively. The spatial resolution of this dataset reaches 17.2 m/pixel. As shown in Fig. 8(e), there exists a ship target with 19 pixels that need to be detected, making up 0.08% of the total amount of pixels in the dataset.

6) Salians Dataset: The sixth dataset was shot by the AVIRIS airborne sensor from Salinas Valley, CA, USA [44]. At a spatial resolution of 3.7 m/pixel, a subimage with a size of 120 × 120 pixels was cropped from the 512 × 217 pixels original image in the experiments. The 25 randomly selected anomalous pixels are synthetic data according to the identified building spectral. After removing 20 atmospheric and water vapor absorption bands (108–112, 154–167, and 224), there are 204 reserved spectral bands ranging from 400 to 2500 nm in the dataset. The pseudo-color image of the Salians dataset is presented in Fig. 8(f).

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B. Experimental Settings

1) Evaluation Indicators: To verify the performance of each algorithm, we implement multiple evaluation indicators to test the detection effectiveness and efficiency of the compared HAD algorithms. First, the running time of each model on six HSI datasets is selected as the evaluation indicator for detection efficiency. In terms of detection effectiveness, we employ multiple evaluation indicators to validate the detection performance from the

perspective of qualitative and quantitative. With respect to the qualitative indicators, we introduce the AD map, background-anomaly separation map [93], and 3-D receiver operating characteristic (ROC) curve [94] to compare multiple state-of-the-art algorithms in our experiments. Specifically, the AD map can directly display the impression of AD and background suppression (BS) from the shade of the color map. By outlining the box plot of the background and anomalies, the background-anomaly separation map can intuitively illustrate the distribution relationship between the background part and the anomaly part. The gap between the anomaly box and the background box reflect the separation degree between the background and the anomaly component in the detection map. In addition, the 3-D ROC curve represents the variation curve of detection probability rate P_d and FA F_a with the variation of segmentation threshold $\tau \in (0, 1)$. Given a segmentation threshold, P_d and P_f are denoted as follows:

$$P_d = \frac{\mathrm{DT}}{\mathrm{AT}}, \quad F_a = \frac{\mathrm{DF}}{\mathrm{AP}} \tag{12}$$

where AT and DT denote the total number of true anomaly pixels in the HSI and the true anomaly pixels detected by the corresponding algorithm, respectively. AP and DF denote the number of all pixels in the HSI and the number of falsely detected pixels under a certain threshold τ. Moreover, the comprehensive 3-D ROC curve evaluation indicator can describe the capability of each method regarding AD and BS. Deriving from the 3-D ROC curve, we can obtain a more quantitative evaluation indicators for algorithm testing experiments. For example, by pinning one of the three indicators P_D, P_D, or τ in 3-D ROC curve, we can derive the 2-D ROC curves for (P_d, F_a), (P_d, τ), and (F_a, τ), respectively. Therefore, by the integral calculation of the corresponding area under the curve (AUC), the quantitative evaluation indicators $\mathrm{AUC}_{(P_d,F_a)} \in (0, 1)$, $\mathrm{AUC}_{(P_d,\tau)} \in (0, 1)$, and $\mathrm{AUC}_{(F_a,\tau)} \in (0, 1)$, can be naturally obtained.

Combining these three evaluation indicators, we can further obtain the following indicators, which reflect the capability of AD, BS, joint AD and BS (ADBS), overall detection (OD), and the signal-to-noise probability ratio (SNPR). The mathematical formula of these indicators are expressed as follows:

$$\mathrm{AUC}_{\mathrm{ad}} = \mathrm{AUC}_{(P_d,F_a)} + \mathrm{AUC}_{(P_d,\tau)} \tag{13}$$

$$\mathrm{AUC}_{\mathrm{bs}} = \mathrm{AUC}_{(P_d,F_a)} - \mathrm{AUC}_{(F_a,\tau)} \tag{14}$$

$$\mathrm{AUC}_{\mathrm{adbs}} = \mathrm{AUC}_{(P_d,\tau)} - \mathrm{AUC}_{(F_a,\tau)} \tag{15}$$

$$\mathrm{AUC}_{\mathrm{od}} = \mathrm{AUC}_{(P_d,F_a)} + \mathrm{AUC}_{(P_d,\tau)} - \mathrm{AUC}_{(F_a,\tau)} \tag{16}$$

$$\mathrm{AUC}_{\mathrm{snpr}} = \frac{\mathrm{AUC}_{(P_d,F_a)}}{\mathrm{AUC}_{(P_d,\tau)}} \tag{17}$$

TABLE III
PARAMETER SETTINGS OF THE TEN COMPARISON HAD ALGORITHMS ON THE CAT ISLAND DATASET

Algorithms	Parameters
OCF-ETLR	$m = 20$, $(\lambda, \alpha, \beta) = (1e^{-3}, 1e^{-3}, 1e^{-1})$
2S-GLRT	$(w_{in}, w_{out}) = (15, 7)$
CRDBPSW	$(w_{in}, w_{out}) = (15, 7)$, $\lambda = 10^{-6}$
EAS-RX	$p = 20$, $j = 8$
LREN	$\lambda = 1e^{-2}$
GAED	$w = 7$, $\beta = 1$
WeaklyAD	Eps $= 0.15$, MinPts $= 1$
DCAE	$\lambda = 1e^{-2}$
BockNet	$B_s = 7$
G2LSSAT	$k = 4$, $n_g = 1$, $n_h = 1$, $l_r = 1e^{-3}$

where higher $\mathrm{AUC}_{\mathrm{ad}} \in [0, 2]$, $\mathrm{AUC}_{\mathrm{bs}} \in [-1, 1]$, $\mathrm{AUC}_{\mathrm{adbs}} \in [-1, 1]$, $\mathrm{AUC}_{\mathrm{od}} \in [-1, 2]$, and $\mathrm{AUC}_{\mathrm{snpr}} \in [0, +\infty]$ values represent better detection performance for AD and BS.

2) Comparison Methods: To investigate the effectiveness and superiority of G2LSSAT, we select four excellent traditional algorithms, including OCF-ETLR [63], 2S-GLRT [17], CRDBPSW [43], EAS-RX [33], and five DNN-based HAD methods, including LREN [78], GAED [65], WeaklyAD [76], DCAE [83], and BockNet [86], to serve as comparison methods in the experiments. As expressed in Table III, we present the key parameters and corresponding settings of these ten comparison methods on the Cat Island dataset. First, for the statistical theory-based method EAS-RX, the parameters of p and j are estimated in accordance with the NWHFC and MXSVD algorithms in [37]. Meanwhile, the parameters of the outer window size w_{out} and inner window size w_{in} in 2S-GLRT are set in the range $\{5, 7, 9, 11, 13, 15, 17, 19\}$ and the range $\{3, 5, 7, 9, 11, 13, 15, 17\}$ to investigate the best performance on different datasets. Then, for the data representation-based algorithm CRDBPSW, the outer window size w_{out} and inner window size w_{in} are selected from the same range as 2S-GLRT. Moreover, the parameter settings of the weighting factor λ is fixed at 10^{-6}. Moreover, OCF-ETLR is an excellent data factorization-based method which relies on low-rank and sparse tensor factorization. In OCF-ETLR, the parameters λ, α, and β are tuned in the range $\{1e^{-5}, 1e^{-4}, 1e^{-3}, 1e^{-2}, 1e^{-1}, 1\}$. The number of bands m is investigated from 5 to 65. For GAED, the window size is varied from 3 to 13 for each dataset. Meanwhile, the penalty coefficient β is selected from the range $\{1e^{-6}, 1e^{-5}, 1e^{-4}, 1e^{-3}, 1e^{-2}, 1e^{-1}, 1, 10\}$. For Auto-AD and DCAE detectors, the tradeoff factor λ is elaborately selected from the range $\{1e^{-5}, 1e^{-4}, 1e^{-3}, 1e^{-2}, 1e^{-1}\}$. For WeaklyAD, the radius value Eps varies from 0.01 to 0.25 for optimal performance, and the value MinPts is tuned from 1 to 25. For BockNet, the number of spots B_s is varied from 1 to 9 for the optimum performance in light of the algorithm literature. For the proposed G2LSSAT, based on python 3.7.0 and the DL framework of PyTorch 1.12.1, we train the network model with the adaptive moment estimation (ADAM) optimizer. The overall training procedure and basic parameter settings of the proposed G2LSSAT are presented in Algorithm 1. The training epoch and batch size are set to 150 and 256, respectively.

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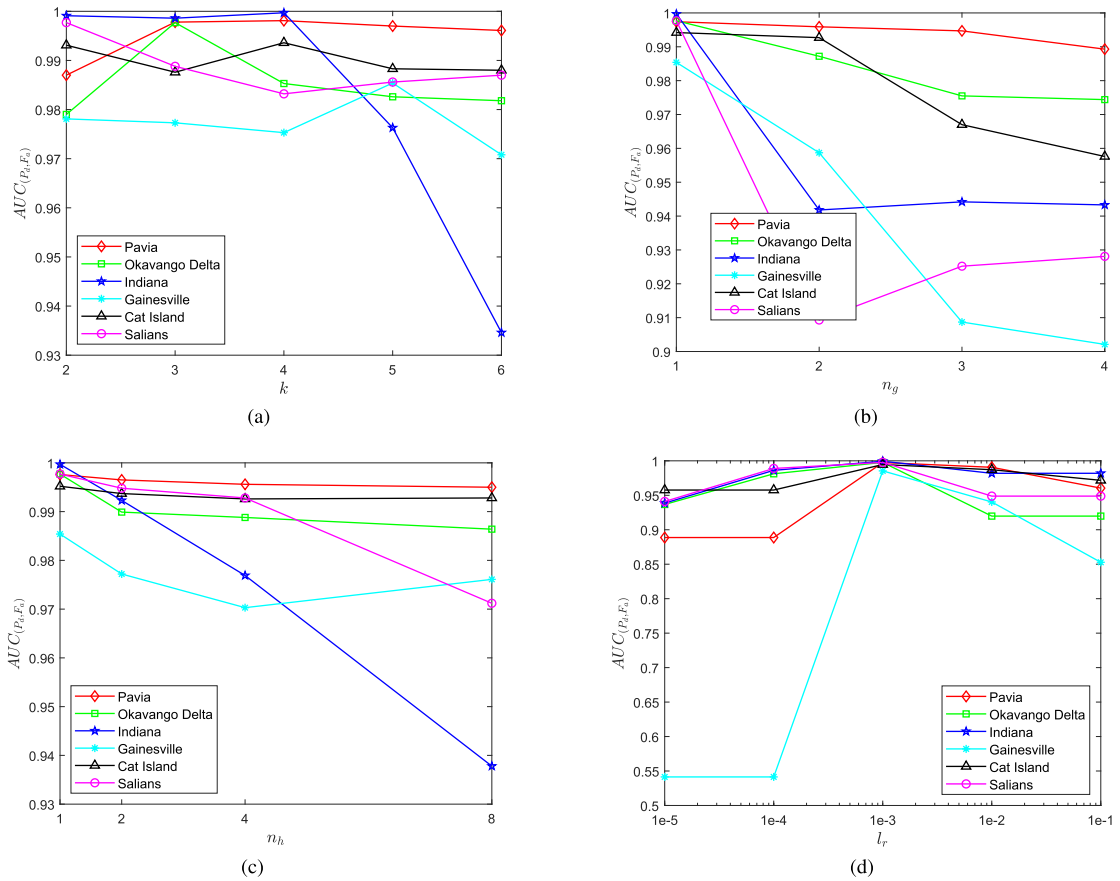


Fig. 9. Analysis of the parameters on the six datasets. (a) Sliding window size k. (b) Number of stacking times n_g of crossbar-based global sparse attention in the GTB. (c) Number of multihead n_h. (d) Learning rate l_r.

In addition, the detailed parameter settings of k, n_g, n_h, and l_r of G2LSSAT on six datasets will be further discussed in Section III-C. Except for LREN, which was implemented based on TensorFlow 2.3.0, the remaining DNN-based models were implemented using PyTorch 1.12.1.

C. Model Parameter Tuning

In this section, we investigate four momentous parameters tuning in the G2LSSAT model, namely, k, n_g, n_h, and l_r. k represents the window size used in the patch partition. l_r denotes the learning rate during the network training. n_g and n_h represents the stacking times and the number of multihead in the crossbar-based global sparse attention module.

1) Impact of k: k is a prominent parameter in the proposed G2LSSAT model, which controls the size of the patch that relates to local feature extraction and background reconstruction. In the experiments, to investigate the effect of sliding size k on the AD performance, we preset k in the range of $\{2, 3, 4, 5, 6\}$ and record the $\text{AUC}_{(P_d,F_a)}$ values on six datasets. It can be seen from Fig. 9(a) that the highest $\text{AUC}_{(P_d,F_a)}$ values are obtained when $k = 4$ on the Pavia and Cat Island datasets. Meanwhile, the highest $\text{AUC}_{(P_d,F_a)}$ values are obtained when $k = 2$ on the Indiana and Salians datasets. Meanwhile, for the Okavango Delta and Gainesville, the optimal values of $\text{AUC}_{(P_d,F_a)}$ are generated when $k = 3$ and $k = 5$, respectively. Therefore, the parameter settings of the sliding size are determined according to the corresponding optimal results. For the multiscale anomalies in these six datasets, the proposed G2LSSAT needs elaborate parameter tuning for the respective sliding windows size k which determines the size of the local receptive field.

2) Impact of n_g: The parameter n_g represents the stacking times of the crossbar-based global sparse attention module in the GTB. We conduct a parameter tuning experiment to investigate whether more attention module stacking can implicitly improve the detection performance of the G2LSSAT. As described in Fig. 9(b), we select the parameter n_g from the range $\{1, 2, 3, 4\}$. It can be seen that the $\text{AUC}_{(P_d,F_a)}$ values gradually decrease when n_g varies from 1 to 4 on all datasets. This funding illustrates that larger n_g cannot bring more vital information but instead the reduces the detection ability. Based on this phenomenon, the parameter n_g is fixed at 1 for the proposed G2LSSAT.

3) Impact of n_h: The design of the multihead is aimed at increasing the flexibility and capability of the transformer model, enabling it to capture more complex feature relationships. Therefore, we also attempt to incorporate multihead self-attention into our method. The $\text{AUC}_{(P_d,F_a)}$ metrics for different n_h are reported in Fig. 9(c). We select the optimal parameter n_g from the range $\{1, 2, 4, 8\}$. From Fig. 9(c), when $n_h = 1$, $\text{AUC}_{(P_d,F_a)}$ is the highest. It indicates that this multihead design cannot be adequately coupled with the GTB

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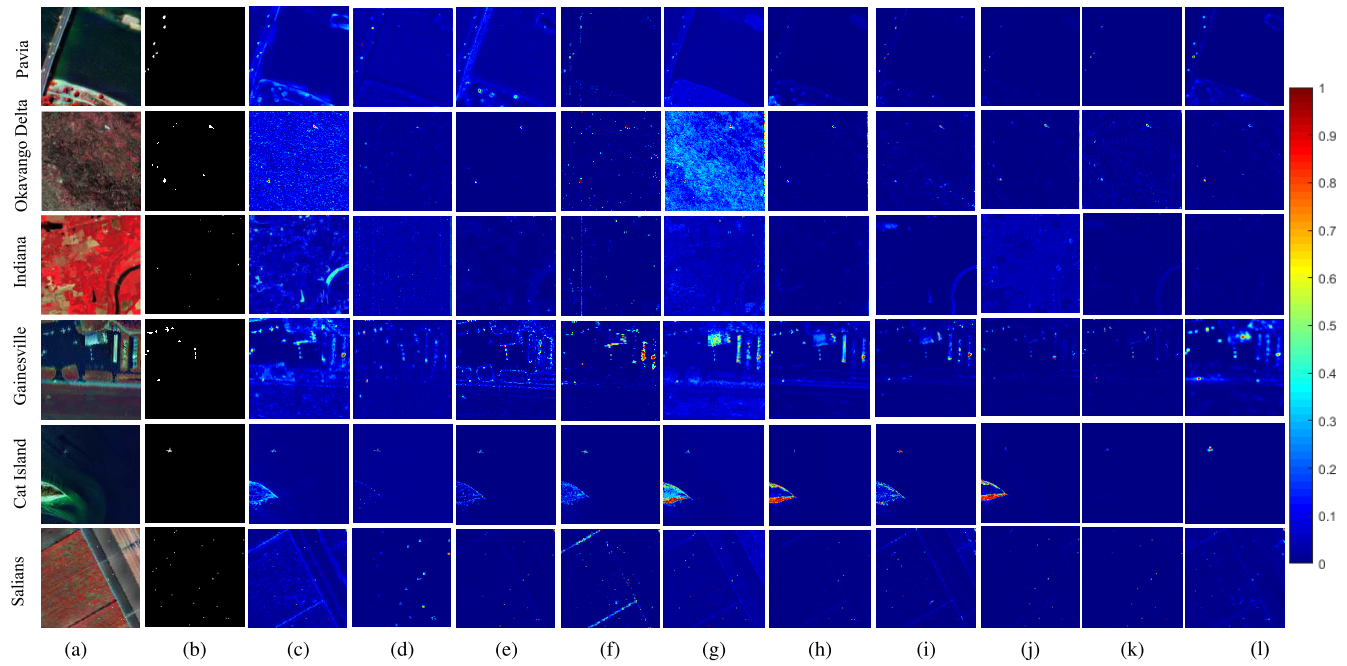


Fig. 10. Color anomaly response maps obtained by ten detectors on six datasets. (a) Pseudo-color image. (b) Ground truth. (c) OCF-ETLR. (d) 2S-GLRT. (e) CRDBPSW. (f) EAS-RX. (g) LREN. (h) GAED. (i) WeaklyAD. (j) DCAE. (k) BockNet. (l) G2LSSAT.

structure. Therefore, in our article, we utilize only the single-head self-attention structure and the parameter of n_h is fixed at 1 during network training and testing.

4) Impact of l_r: The learning rate l_r is an important hyper-parameter that determines whether the objective loss function can converge to a local minimum and when it will converge to the minimum. An appropriate learning rate can enable the objective function to converge around a local minimum within an appropriate training time and then improve the detection performance of the network. The parameters of learning rate l_r are tuned in the range $\{1e^{-5}, 1e^{-4}, 1e^{-3}, 1e^{-2}, 1e^{-1}\}$. Fig. 9(d) lists the $\text{AUC}_{(P_d,F_a)}$ indicators for different learning rates on all datasets. From Fig. 9(d), we can observe that when $l_r = 1e^{-3}$, our G2LSSAT yields the highest $\text{AUC}_{(P_d,F_a)}$ on the all datasets. Consequently, in our experiment, we set l_r as $1e^{-3}$ to accurately detect anomalies.

D. Model Detection Accuracy Comparison

1) Qualitative Comparison Results: As presented in Fig. 10, we show the visual color anomaly response maps of the ten comparison methods on six HSI datasets. The anomaly response map can clearly describe the response intensity of anomalies among the HAD methods. It is noticed that the response map in Fig. 10 varies from deep blue to deep red, which corresponds to the color bar and demonstrates that the anomaly response intensity is growing. First, the anomaly response maps of these ten detection algorithms on the Pavia, Okavango Delta, Indiana, and Gainesville datasets are presented from the first row to the fourth row in Fig. 10. It can be observed that the CRDBPSW, GAED, WeaklyAD, LREN, and BockNet models demonstrate low responses to anomalies, resulting in a certain of missed detections (MDs) and incomplete target shape reconstruction. The anomaly targets can be highlighted from the response maps by the OCF-ETLR, EAS-RX, 2S-GLRT, and DCAE algorithms.

However, some pronounced FAs are produced due to defects in the BS ability. In contrast, our G2LSSAT has the superiority in both detecting anomalies and suppressing background elements. In addition, the small anomalies in the Cat Island and Salians datasets, the anomaly response maps of these ten algorithms are illustrated in the last two rows of Fig. 10. It can be observed that the OCF-ETLR, EAS-RX, LREN, GAED, and WeaklyAD algorithms generate a series of FAs pixels in these two scenes, which obviously contaminates the detection results. Moreover, the 2S-GLRT, DCAE, BockNet, and CRDBPSW struggle to accurately detect all anomalies in the Salians dataset. Overall, the proposed G2LSSAT achieves a more discriminative response to anomaly targets compared to the comparison algorithms.

In addition, the ROC curve for (P_d, F_a), (P_d, τ), (F_a, τ), and (P_d, F_a, τ) of the nine HAD algorithms on six datasets are illustrated in Fig. 11. First, it can be observed from the ROC curve for (P_d, F_a) on the six datasets that the proposed the G2LSSAT model is closest to the upper left corner of the curve, which indicates the superiority of its comprehensive detection performance compared to that of the other methods. Then, as depicted in Fig. 11, the 2-D ROC curve of (P_d, τ) obtained by the proposed method is proximate to the upper-right corner of the Pavia, Gainesville, and Cat Island datasets. The proximity of the 2-D (P_d, τ) ROC curve to the upper-right corner is the directive of the proposed G2LSSAT obtaining superior anomaly discovery performance. Similarly, the 2-D ROC curve for (F_a, τ) of the proposed G2LSSAT can quickly reach the lower-left corner of the coordinate axis on the Okavango Delta and Indiana datasets. This phenomenon demonstrates that the background suppressing performance of G2LSSAT is satisfactory on these two datasets.

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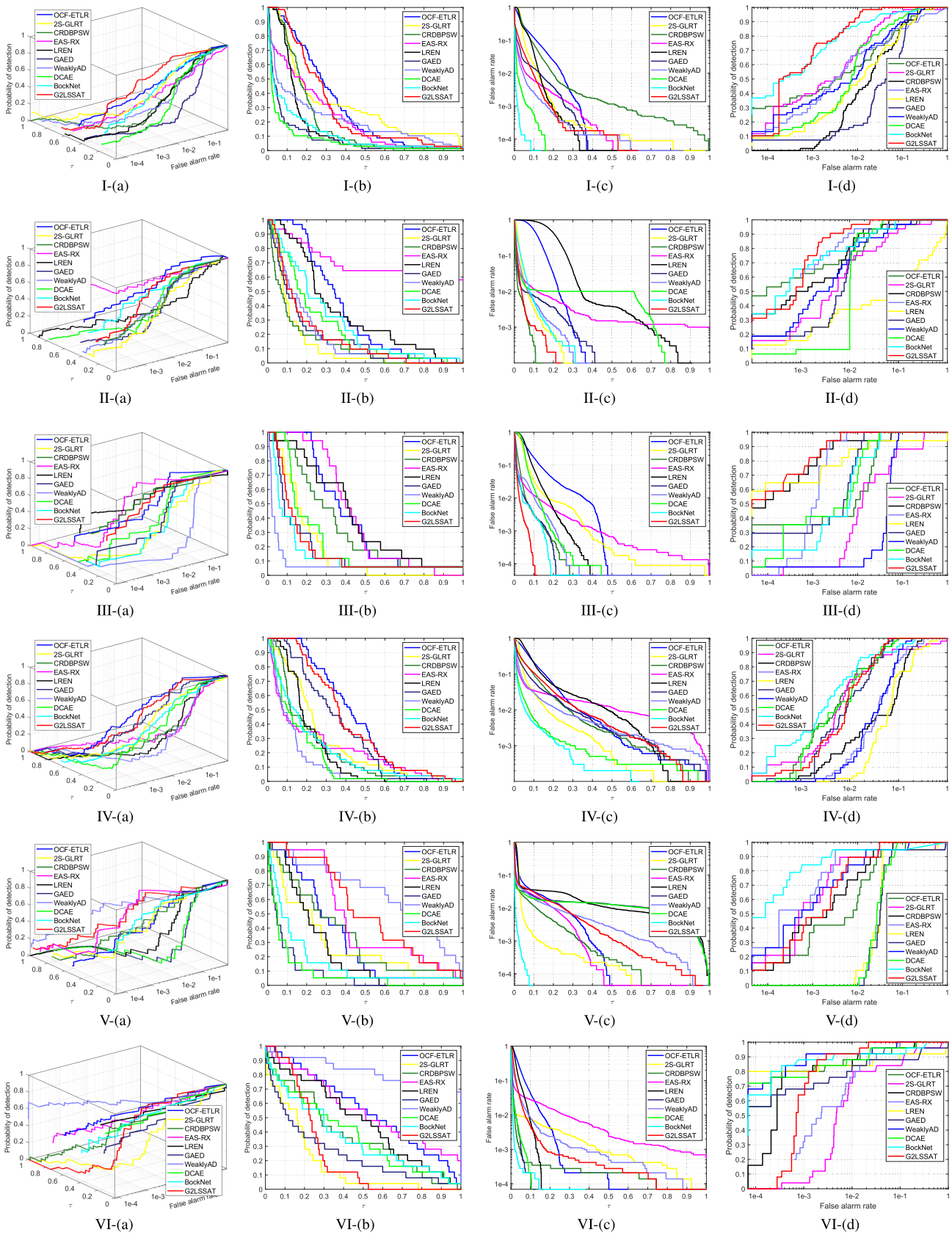


Fig. 11. Three-dimensional (P_d, F_a, τ) ROC curves, 2-D (P_d, τ) ROC curves, 2-D (F_a, τ) ROC curves, and 2-D (P_d, F_a, τ) ROC curves of ten HAD methods on six datasets. (I-a)–(I-d) Pavia. (II-a)–(II-d) Okavango Delta. (III-a)–(III-d) Indiana. (IV-a)–(IV-d) Gainesville. (V-a)–(V-d) Cat Island. (VI-a)–(VI-d) Salians.

Moreover, as depicted in Fig. 12, we utilize box-whisker plots to represent anomaly-background separation maps for intuitively reporting the distribution discrimination between the anomalies and the background in the detection maps.

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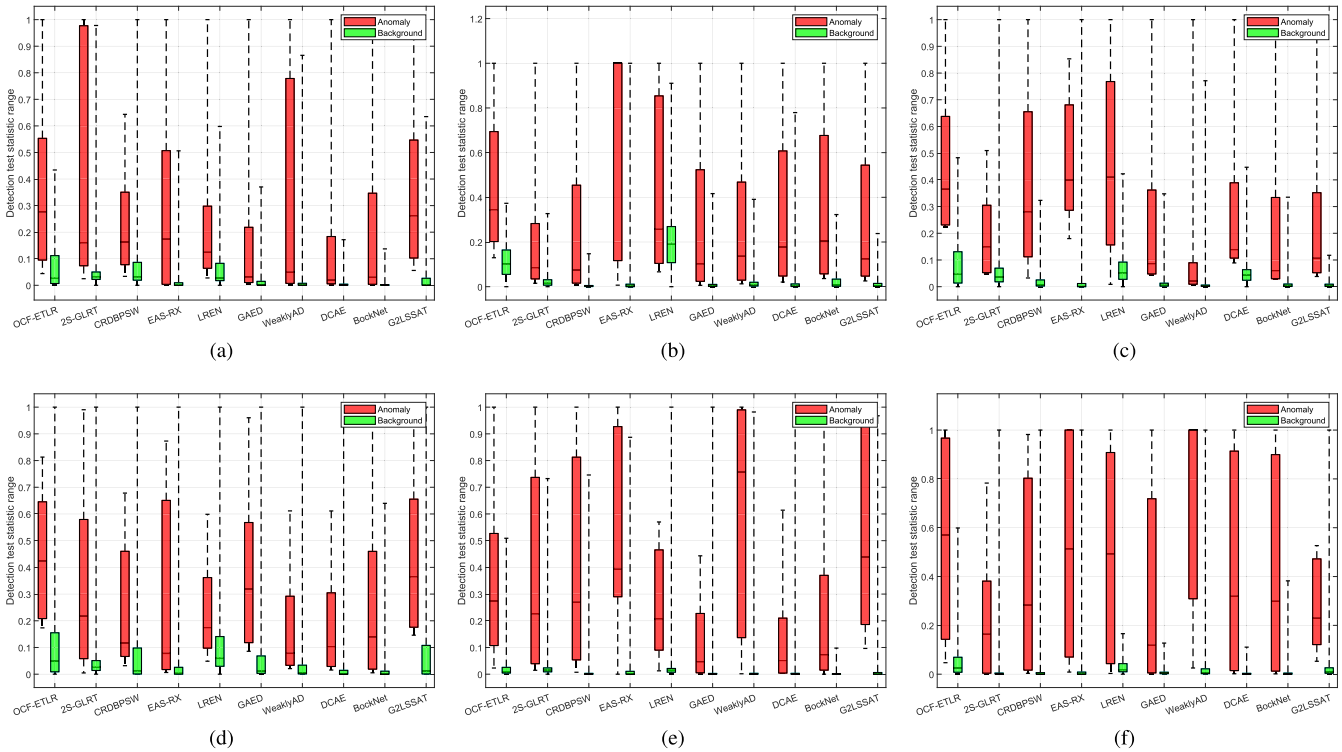


Fig. 12. Anomaly-background separation maps of the ten detectors on the six datasets. (a) Pavia. (b) Okavango Delta. (c) Indiana. (d) Gainesville. (e) Cat Island. (f) Salians.

In Fig. 12, the red box and blue box denote the abnormal pixel distribution and background pixel distribution, respectively. Specifically, the larger gap between these two boxes represents better anomalies and background separability of the algorithm. It can be observed from Fig. 12 that the proposed G2LSSAT gains pretty well performance in separation between the background and anomalies on the six HSI datasets especially on the Pavia and Gainesville datasets, which demonstrate the excellent capability of G2LSSAT for anomalies and background separation compared to other detectors.

2) Quantitative Comparison Results: Except for the qualitative comparison results, we report the quantitative comparison results of each algorithm to quantitatively evaluate the detection effect. As recorded in Table IV, we present the eight evaluation indicators of the AUC according to the 3-D ROC curve, with the optimal results highlighted in bold and the second-best results are underlined. It can be observed from Table IV that our G2LSSAT yields the highest values of $\mathrm{AUC}_{(P_d,F_a)}$, $\mathrm{AUC}_{\mathrm{adbs}}$, and $\mathrm{AUC}_{\mathrm{od}}$, on multiple datasets such as the Pavia and Gainesville datasets, which demonstrates the superiority of G2LSSSAT in terms of OD performance. Except for the Gainesville dataset, the proposed G2LSSAT model gains optimal $\mathrm{AUC}_{(P_d,F_a)}$ on other datasets. Specifically, for the values of $\mathrm{AUC}_{(P_d,F_a)}$, the proposed G2LSSAT yields an essential improvement of 0.78%, 0.51%, 0.08%, 0.31%, and 0.23% on the other five datasets compared to the second-best detector. Meanwhile, the proposed method also obtains the exceptional $\mathrm{AUC}_{\mathrm{snpr}}$ on the Cat Island, Okavango Delta, Indiana, and Pavia datasets, which directly proves the reasonable background suppressing performance of our G2LSSAT. Consequently, considering these qualitative and quantitative comparison results comprehensively, it can be concluded that our G2LSSAT achieves the most robust and second-to-none HAD accuracy performance among the comparison detectors.

The proposed G2LSSAT model showcases clear superiority in detecting multiple types of anomalies spanning different scenarios, as revealed by numerous evaluation indicators results derived from the above qualitative and quantitative comparison experiments. In comparison to the compared DNN-based models (LREN, GAED, WeaklyAD, DCAE, and BockNet), the proposed G2LSSAT almost obtains optimal detection accuracy performance about evaluation indicators on all datasets. To our knowledge, this is mainly attributed to the fact that the proposed G2LSSAT incorporates a global-to-local attention mechanism and invisible diagonal masking scheme, which fully explores the global and local spatial–spectral correlation features and overshadows the reconstruction of anomalies, making the reconstructed image further away from anomalies while closer to the real background.

E. Model Running Time Comparison

As illustrated in Table V, we report the running time of ten detectors on all datasets to investigate the computational burden of each algorithm. The WeaklyAD, DCAE, BockNet, and G2LSSAT are implemented based on the PyTorch and trained by the accelerated computing of GPU. The LREN is implemented based on the TensorFlow by means of the accelerated computing of GPU. The GAED model and four traditional models are also implemented with the Python environment. Among the four traditional detectors, EAS-RX expends the shortest running time in comparison to other

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TABLE IV
AUC VALUES OF TEN DETECTORS THE SIX DATASETS

Data Sets	AUC vaule	OCF-ETLR	2S-GLRT	CRDBPSW	EAS-RX	LREN	GAED	WeaklyAD	DCAE	BockNet	G2LSSAT
Pavia	$AUC_{(P_d,F_a)}\uparrow$	0.9637	0.9772	0.9506	0.9367	0.9521	0.9374	0.9680	0.9782	0.9898	**0.9976**
	$AUC_{(P_d,\tau)}\uparrow$	0.3151	**0.3222**	0.1927	0.2277	0.1691	0.0916	0.2165	0.0805	0.1238	0.2985
	$AUC_{(F_a,\tau)}\downarrow$	0.0445	0.0364	0.0452	0.0047	0.0388	0.0080	0.0038	0.0020	**0.0007**	0.0086
	$AUC_{ad}\uparrow$	1.2789	**1.2994**	1.1433	1.1645	1.1213	1.0290	1.1845	1.0588	1.1136	1.2962
	$AUC_{bs}\uparrow$	0.9192	0.9408	0.9054	0.9321	0.9134	0.9294	0.9642	0.9762	**0.9892**	0.9890
	$AUC_{adbs}\uparrow$	0.2707	0.2857	0.1474	0.2231	0.1304	0.0836	0.2127	0.0785	0.1231	**0.2899**
	$AUC_{od}\uparrow$	1.2344	1.2629	1.0981	1.1598	1.0825	1.0209	1.1806	1.0568	1.1129	**1.2876**
	$AUC_{snpr}\uparrow$	7.0850	8.8438	4.2584	48.7830	4.3621	11.4012	56.7126	40.0129	**186.9758**	34.7502
Okavango Delta	$AUC_{(P_d,F_a)}\uparrow$	0.9838	0.9676	0.9926	0.9902	0.7308	0.9838	0.9836	0.9872	0.9908	**0.9977**
	$AUC_{(P_d,\tau)}\uparrow$	0.3765	0.1353	0.1442	**0.7077**	0.3582	0.1907	0.1893	0.2482	0.2923	0.1980
	$AUC_{(F_a,\tau)}\downarrow$	0.1070	0.0178	**0.0030**	0.0082	0.1902	0.0065	0.0102	0.0117	0.0146	0.0069
	$AUC_{ad}\uparrow$	1.3603	1.1029	1.1369	**1.6979**	1.0890	1.1745	1.1729	1.2354	1.2831	1.1957
	$AUC_{bs}\uparrow$	0.8767	0.9498	0.9896	0.9820	0.5406	0.9773	0.9734	0.9754	0.9763	**0.9909**
	$AUC_{adbs}\uparrow$	0.2695	0.1176	0.1412	**0.6995**	0.1680	0.1842	0.1791	0.2365	0.2777	0.1911
	$AUC_{od}\uparrow$	1.2532	1.0852	1.1339	**1.6897**	0.8988	1.1680	1.1627	1.2237	1.2686	1.1888
	$AUC_{snpr}\uparrow$	3.5177	7.6182	48.1898	**86.5048**	1.8832	29.5206	18.5655	21.201	20.0692	28.8156
Indiana	$AUC_{(P_d,F_a)}\uparrow$	0.9886	0.9478	0.9963	0.9986	0.9395	0.9950	0.9532	0.9931	0.9915	**0.9994**
	$AUC_{(P_d,\tau)}\uparrow$	0.4244	0.1825	0.3542	**0.4355**	0.4348	0.1986	0.0893	0.2345	0.1740	0.2090
	$AUC_{(F_a,\tau)}\downarrow$	0.0639	0.0427	0.0117	0.0067	0.0576	0.0075	0.0066	0.0451	0.0054	**0.0045**
	$AUC_{ad}\uparrow$	1.4130	1.1303	1.3505	**1.4341**	1.3743	1.1936	1.0425	1.2276	1.1655	1.2084
	$AUC_{bs}\uparrow$	0.9247	0.9051	0.9846	0.9919	0.8819	0.9875	0.9466	0.9480	0.9861	**0.9949**
	$AUC_{adbs}\uparrow$	0.3605	0.1398	0.3425	**0.4288**	0.3772	0.1911	0.0827	0.1894	0.1686	0.2045
	$AUC_{od}\uparrow$	1.3491	1.0876	1.3388	**1.4274**	1.3166	1.1861	1.0359	1.1825	1.1601	1.2039
	$AUC_{snpr}\uparrow$	6.6432	4.2721	30.3026	**65.0097**	7.5457	26.5136	13.5481	5.1990	32.3117	46.4585
Gainesville	$AUC_{(P_d,F_a)}\uparrow$	**0.9860**	0.9444	0.9249	0.9388	0.8966	0.9838	0.9512	0.9852	0.9816	0.9854
	$AUC_{(P_d,\tau)}\uparrow$	**0.4272**	0.2704	0.2060	0.2182	0.2089	0.3485	0.1229	0.1520	0.2124	0.4074
	$AUC_{(F_a,\tau)}\downarrow$	0.0695	0.0324	0.0357	0.0215	0.0805	0.0296	0.0179	0.0067	**0.0053**	0.0383
	$AUC_{ad}\uparrow$	**1.4131**	1.2148	1.1308	1.1570	1.1055	1.3323	1.0741	1.1371	1.1940	1.3929
	$AUC_{bs}\uparrow$	0.9165	0.9120	0.8892	0.9174	0.8160	0.9542	0.9333	**0.9785**	0.9763	0.9472
	$AUC_{adbs}\uparrow$	0.3577	0.2381	0.1703	0.1967	0.1284	0.3189	0.1051	0.1453	0.2071	**0.3692**
	$AUC_{od}\uparrow$	1.3437	1.1824	1.0951	1.1356	1.0250	1.3027	1.0563	1.1304	1.1887	**1.3546**
	$AUC_{snpr}\uparrow$	6.1494	8.3524	5.7725	10.1656	2.5941	11.7898	6.8784	22.7068	**40.1623**	10.6453
Cat Island	$AUC_{(P_d,F_a)}\uparrow$	0.9854	0.9685	0.9921	0.9500	0.9419	0.9229	0.9900	0.9680	0.9793	**0.9952**
	$AUC_{(P_d,\tau)}\uparrow$	0.3405	0.2906	0.3472	0.4771	0.2327	0.0852	**0.6532**	0.0904	0.1840	0.5303
	$AUC_{(F_a,\tau)}\downarrow$	0.0189	0.0164	0.0046	0.0088	0.0292	0.0152	0.0078	0.0135	**0.0003**	0.0098
	$AUC_{ad}\uparrow$	1.3259	1.2592	1.3393	1.4271	1.1746	1.0081	**1.6432**	1.0584	1.1632	1.5255
	$AUC_{bs}\uparrow$	0.9665	0.9521	**0.9875**	0.9412	0.9127	0.9077	0.9822	0.9545	0.9790	0.9854
	$AUC_{adbs}\uparrow$	0.3216	0.2742	0.3426	0.4684	0.2035	0.0700	**0.6454**	0.0770	0.1837	0.5205
	$AUC_{od}\uparrow$	1.3070	1.2428	1.3347	1.4183	1.1454	0.9929	**1.6354**	1.0450	1.1630	1.5157
	$AUC_{snpr}\uparrow$	18.0193	17.7102	76.1377	54.2524	7.9730	5.5941	83.9003	6.7209	**691.5452**	54.0161
Salians	$AUC_{(P_d,F_a)}\uparrow$	0.9877	0.9491	0.9891	0.9888	0.9194	0.9374	0.9954	0.9893	0.9812	**0.9977**
	$AUC_{(P_d,\tau)}\uparrow$	0.5744	0.1744	0.3863	0.5572	0.4903	0.2652	**0.8245**	0.3939	0.3723	0.2453
	$AUC_{(F_a,\tau)}\downarrow$	0.0344	0.0049	0.0029	0.0111	0.0239	0.0049	0.0107	0.0018	**0.0017**	0.0135
	$AUC_{ad}\uparrow$	1.5620	1.1235	1.3754	1.5460	1.4097	1.2026	**1.8199**	1.3832	1.3536	1.2430
	$AUC_{bs}\uparrow$	0.9533	0.9441	0.9862	0.9777	0.8955	0.9325	0.9847	**0.9875**	0.9796	0.9842
	$AUC_{adbs}\uparrow$	0.5400	0.1695	0.3834	0.5460	0.4664	0.2603	**0.8138**	0.3920	0.3707	0.2318
	$AUC_{od}\uparrow$	1.5276	1.1185	1.3725	1.5349	1.3858	1.1977	**1.8092**	1.3814	1.3519	1.2295
	$AUC_{snpr}\uparrow$	16.7006	35.3605	131.9843	50.0593	20.5104	53.7820	77.1209	213.8339	**221.7635**	18.1430

TABLE V
RUNNING TIME (IN SECONDS) OF TEN ALGORITHMS ON THE SIX DATASETS. TR AND TE REPRESENT THE RUNNING TIME FOR TRAINING AND TESTING

Data Sets	Model-based Detectors				DNN-based Detectors (Tr/Te)					
	OCF-ETLR	2S-GLRT	CRDBPSW	EAS-RX	LREN	GAED	WeaklyAD	DCAE	BockNet	G2LSSAT
Pavia	7.133	46.286	36.216	**2.879**	**15.890**/104.480	121.177/**0.217**	375.264/0.251	25.656/0.173	100.998/1.027	114.767/0.405
Okavango Delta	11.515	24.775	25.612	**1.817**	70.290/48.800	58.234/0.078	78.392/0.175	**8.363/0.021**	118.525/2.806	64.394/0.170
Indiana	23.648	57.544	55.924	**4.72**	152.710/109.050	136.564/0.183	132.073/0.323	**23.534/0.094**	238.490/0.481	166.840/0.463
Gainesville	14.350	29.660	11.034	**4.331**	38.690/48.220	65.233/0.081	194.800/0.181	**8.798/0.023**	102.973/0.952	88.872/0.283
Cat Island	23.676	78.439	38.551	**8.236**	**15.080**/106.850	136.012/0.189	388.804/0.375	25.302/**0.108**	100.957/0.985	205.683/0.418
Salians	16.112	65.565	38.332	**6.863**	105.380/68.360	101.567/0.169	208.444/0.252	**14.737/0.045**	182.790/3.379	77.402/0.244

methods, and it performs comparatively well in detection accuracy on the Okavango Delta and Indiana datasets. It can be seen from Table V that the six DNN-based detectors require a certain amount of training time, among which DCAE expends

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TABLE VI
ABLATION STUDY OF THE PROPOSED METHOD ON THE SIX DATASETS

Data Sets	Comparison Model					
	baseline	baseline+GTB	baseline+LTB	baseline+LTB+IDM	baseline+GTB+LTB	baseline+GTB+LTB+IDM
Pavia	0.9929	0.9944	0.9958	0.9961	0.9965	**0.9976**
Okavango Delta	0.9767	0.9819	0.9843	0.9902	0.9889	**0.9977**
Indiana	0.9756	0.9825	0.9856	0.9935	0.9947	**0.9994**
Gainesville	0.9037	0.9673	0.9644	0.9703	0.9790	**0.9854**
Cat Island	0.9766	0.9856	0.9896	0.9926	0.9937	**0.9952**
Salians	0.9699	0.9817	0.9926	0.9951	0.9959	**0.9977**
Average	0.9659	0.9822	0.9853	0.9914	0.9899	**0.9955**

the shortest training time on multiple datasets due to its simpler network architecture and fewer network parameters. In addition, apart from LREN requiring more running time on dictionary construction and matrix factorization, the other five DNN-based models all present extremely few network inference times on six HSI datasets, while our G2LSSAT takes relatively less inference time on different datasets. In summary, our G2LSSAT obtains a relatively reasonable tradeoff between detection accuracy and efficiency. Note that, since the proposed G2LSSAT adopts joint local patches-based and global crossbar-based sampling methods in the training stage, it undeniably consumes extra training and inference time compared to other reconstruction-based detectors such as DCAE and GAED.

TABLE VII
COMPARATIVE PERFORMANCE OF GTB AND MHGA OVER $AUC_{(P_d,F_a)}$ AND RUNNING TIME ON THE SIX DATASETS. MHGA AND GTB REPRESENT THE MHGA MODULE AND THE GTB MODULE IN OUR METHOD

Data Sets	$AUC_{(P_d,F_a)}$		Running Time(s)	
	MHGA	GTB	MHGA	GTB
Pavia	0.9981	0.9976(-0.0005)	195.74	114.76(-80.98)
Okavango Delta	0.9980	0.9977(+0.0003)	86.86	64.39(-22.47)
Indiana	0.9993	0.9994(+0.0001)	182.27	166.84(-15.43)
Gainesville	0.9829	0.9854(+0.0025)	112.61	88.87(-23.74)
Cat Island	0.9950	0.9952(+0.0002)	231.91	205.68(-26.23)
Salians	0.9959	0.9977(+0.0018)	96.51	77.40(-19.11)
Average	0.9948	0.9955(+0.0007)	150.98	119.66(-31.32)

F. Ablation Study

To evaluate the influences of three central modules, i.e., GTB, LTB, and IDM, on the proposed G2LSSAT, we implement some ablation studies on six datasets. The detailed experimental results are delineated in Table VI.

1) Effectiveness of GTB: As shown in Table VI, we implement six experimental control groups to investigate the effectiveness of GTB, LTB, and IDM. Note that the baseline in Table VI represents the basic model without the utilization of these three components. First, to evaluate the effect of the GTB on the proposed G2LSSAT, we compare the baseline model with the baseline+GTB model and record the values of $AUC_{(P_d,F_a)}$ on the six datasets. It can be observed from Table VI that the improved model, which embeds the GTB into the baseline model, results in increments of 0.15%, 0.52%, 0.69%, 6.36%, 0.9%, and 1.18%, relative to the baseline model for each respective dataset. This is mainly because the GTB module leverages a crossbar-based global sparse attention mechanism to capture and integrate global information, which augments the learning of nonlocal representations. In addition, as presented in Table VII, we conduct an experimental comparison between the performances of basic multihead global self-attention (MHGA) [95] and the proposed GTB. Specifically, the MHGA will independently represent each pixel by the joint subspace of global spatial correlation information. It can be concluded from Table VII that upon utilization of GTB, the G2LSSAT results in a significant increase in computational efficiency (31.32 s average reduction of running time) and approximate detection accuracy [0.07% average increment of $AUC_{(P_d,F_a)}$] compared to the G2LSSAT model with MHGA, which further proves that the proposed GTB has the ability of global correlation representation learning similar to the MHGA while improving computing efficiency by relying on the elaborate scheme of crossbar-based global sparse attention.

2) Effectiveness of LTB: In the experiments, we follow the global-to-local reconstruction network framework and incorporate the local transformer which extracts the local reconstruction features based on the local patch-based self-attention module. To verify the effectiveness of LTB, we conduct an ablation study of LTB and present the experimental results in Table VI. It can be seen that the $AUC_{(P_d,F_a)}$ values derived from the improved baseline+LTB model yield considerable increases of 0.29%, 0.76%, 1%, 6.07%, 1.3%, and 2.27% compared to the baseline model, which fully indicates that the detection accuracy can be boosted by exploiting the spectral–spatial correlation features of the local background. Meanwhile, the improved baseline+LTB+GTB, which combines the GTB module with the LTB module, can effectively bring about an average increase of 2.43% on $AUC_{(P_d,F_a)}$. This shows that these two components can be well coupled and further improve the detection performance of the proposed G2LSSAT.

3) Effectiveness of IDM: In our method, we design a IDM to suppress the anomalies over-reconstruction issue during network training. To investigate the effectiveness of the IDM on G2LSSAT, several ablation experiments have been quantitatively and qualitatively devised. First, as

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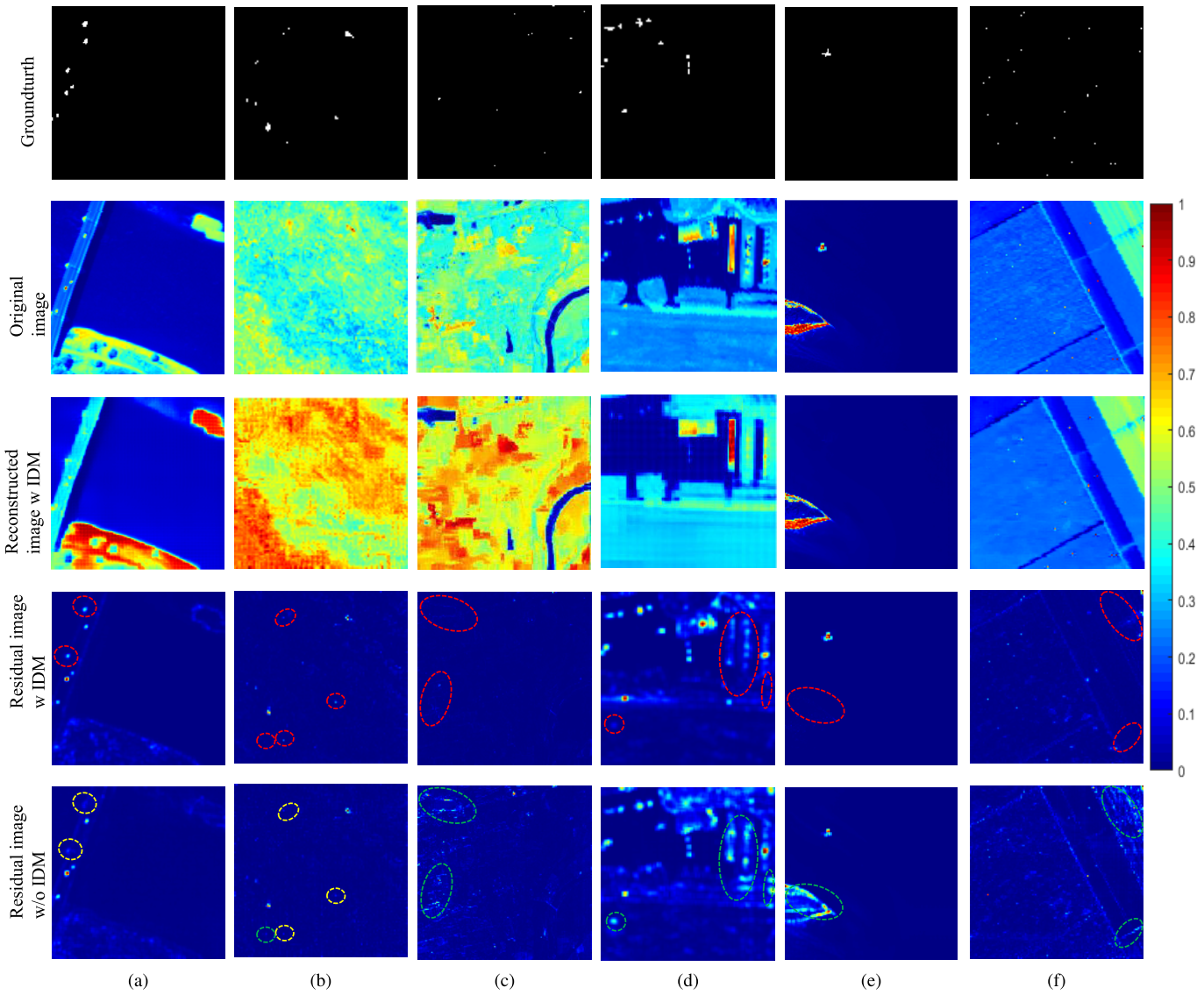


Fig. 13. Ground truth and color anomaly response maps obtained by the proposed G2LSSAT model with IDM and without IDM. The second and third rows in the figure present the original HSIs and reconstructed HSIs with our G2LSSAT. The yellow region and green region circled in the fourth and fifth rows of the figure represent the MDs and FAs of the G2LSSAT model without IDM in comparison to those of the G2LSSAT model. The red regions denote the corresponding regions in the anomaly response map of the G2LSSAT model with IDM. (a) Pavia. (b) Okavango Delta. (c) Indiana. (d) Gainesville. (e) Cat Island. (f) Salians.

shown in Table VI, the baseline+GTB+LTB+IDM and the baseline+GTB+LTB represents the complete the G2LSSAT model and the G2LSSAT model without the IDM. From the quantitative results over $\mathrm{AUC}_{(P_d,F_a)}$, we can see that the $\mathrm{AUC}_{(P_d,F_a)}$ values derived from the G2LSSAT model with IDM are higher than that of the G2LSSAT model with IDM on six HSI datasets. Notably, relative to the controlled baseline+LTB model, the evolved baseline+LTB+IDM model also boosts the detection accuracy performance due to the implementation of the IDM. Then, we present the original HSI and reconstructed HSI with our G2LSSAT in the second and third rows of Fig. 13. As presented in the reconstructed HSIs of Pavia and Cat Island datasets in Fig. 13, it can be visually observed that the anomalies in reconstructed HSI are indeed overwhelmed by the background, which proves the overall anomalies reconstruction suppression effect. Furthermore, as elucidated in the fourth and fifth rows of Fig. 13, we intuitively compare the detection effect of these two models in light of the anomaly response maps. Compared to the complete G2LSSAT model, the absence of the IDM results in the serious occurrences of FAs and MDs, which are triggered by the decrease in background reconstruction residuals and the increase in anomalies reconstruction residuals. This finding confirms that the IDM in the HAD task results in a notable augmentation of anomalies reconstruction inhibition and background reconstruction assistance from the opposite side.

G. Model Complexity and Convergence

1) Complexity Analysis: In this section, we investigate the complexity of our G2LSSAT model. In the spirit of equality, we mainly compare our method with the other two excellent DNN-based reconstruction network, DCAE and BockNet,

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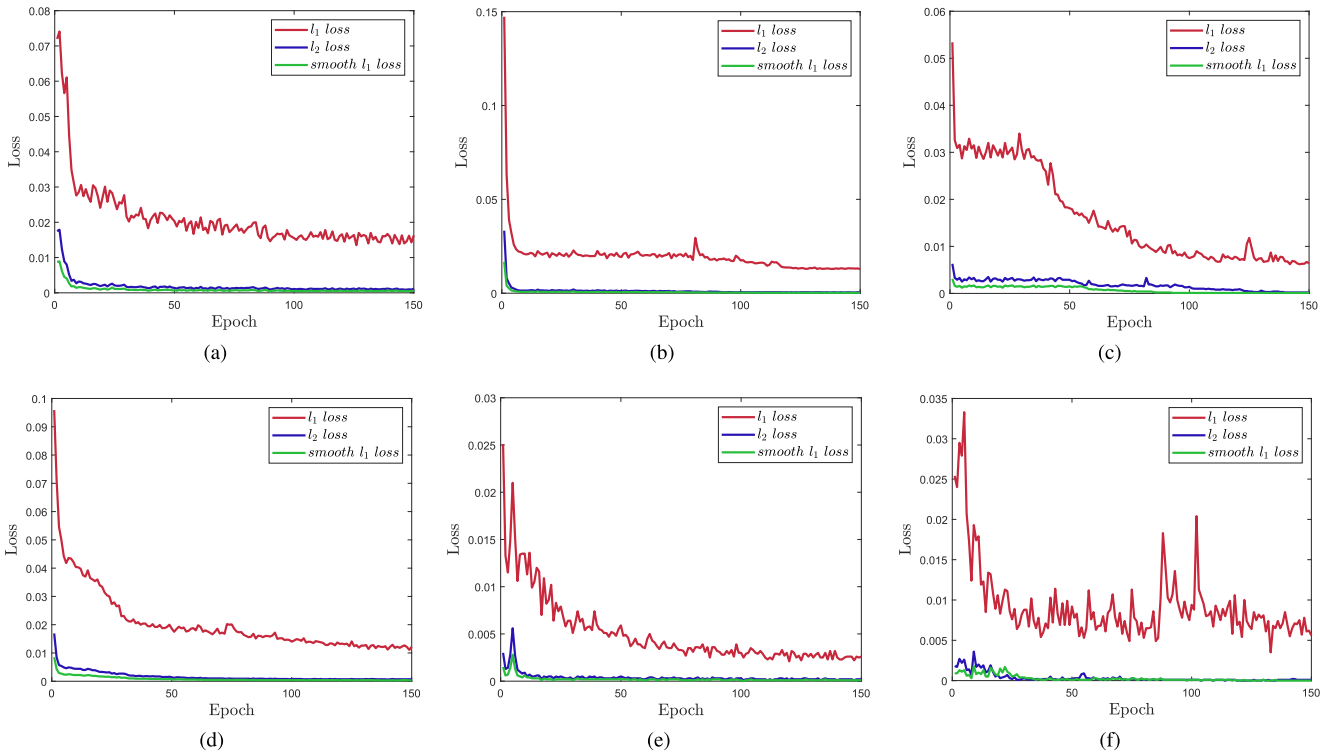


Fig. 14. Comparative performance of the l_1, l_1, and smooth l_1 loss functions over convergence characteristic curves on the six datasets. (a) Pavia. (b) Okavango Delta. (c) Indiana. (d) Gainesville. (e) Cat Island. (f) Salians.

TABLE VIII
MODEL COMPLEXITY ANALYSIS OF THE THREE DNN-BASED MODELS

Indicators	DCAE	BockNet	G2LSSAT
Average Training Time (s)	**17.73**	140.78	119.66
Average Testing Time (s)	**0.08**	1.61	0.33
Params (MB)	**0.11**	0.71	0.43
Average AUC$_{(P_d,F_a)}$	0.9835	0.9857	**0.9955**

based on the same software and hardware implementation platform. The comparison results of network parameter quantity, average training time, average testing time, and average AUC$_{(P_d,F_a)}$ are listed in Table VIII. Compared to DCAE and BockNet, G2LSSAT has the highest AUC$_{(P_d,F_a)}$ and a moderate level of running time and network parameters. Although DCAE has less running time and network parameters than G2LSSAT, the detection accuracy of DCAE has no advantages compared with our method. In summary, the proposed G2LSSAT exhibits promising detection accuracy performance and competitive efficiency advantages.

2) Convergence Analysis: As described in Figs. 14 and 15, we conduct convergence characteristic analysis experiments of the proposed G2LSSAT on all datasets. In our experiments, we introduce three different loss functions, i.e., l_1, l_2, and smooth l_1, to observe the corresponding detection and convergence performances. l_2 loss function stands for the mse loss function. From the loss curves of l_1, l_2, and smooth l_1 loss in Fig. 14, it can be seen that the loss curves of l_2 loss and smooth l_1 loss decrease at a faster rate than that provided by the l_1 loss function. Moreover, among these three loss functions, the l_2 loss function has distinct advantages in

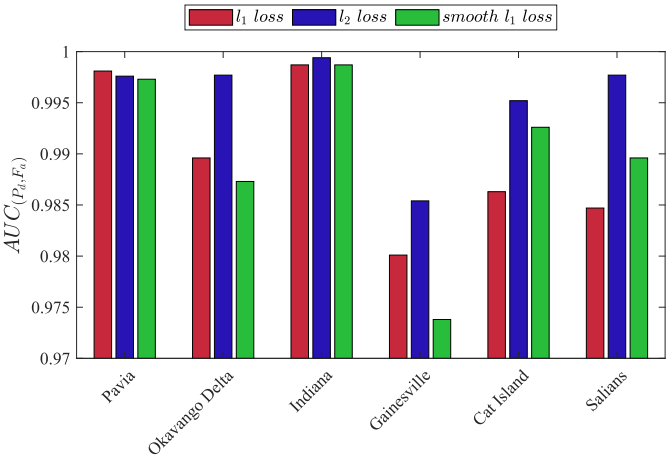


Fig. 15. AUC$_{(P_d,F_a)}$ of the three loss functions for the six datasets.

terms of AUC$_{(P_d,F_a)}$ outcomes, as shown in Fig. 15. These observations, together with the loss curves in Fig. 14, lead us to conclude that the l_2 loss is excellent in terms of detection accuracy and convergence stability for the proposed method.

IV. CONCLUSION

In this article, we devise a transformer-based G2LSSAT network model to achieve robust AD in HSIs. The proposed model employs a global-to-local network structure in light of the visual attention mechanism in HVS. To be specific, the GTB and LTB are introduced in sequence to extract the global and local spatial–spectral reconstruction features. In addition, we embed the elaborate crossbar-based global

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sparse attention and patch-based local self-invisible attention into the GTB and LTB modules to explore the long-range and local relation characteristics. Moreover, to avoid unexpected anomalies reconstruction problem, we design a IDM to erase abnormal pixels itself in the receptive field and effectuate accurate reconstruction by means of the global and local dependent features. Extensive comparison experiments and ablation studies on six datasets demonstrate the superiority and robustness of the proposed G2LSSAT compared with the state-of-the-art detectors. However, the proposed G2LSSAT also inevitably suffers from the flaws of low computational efficiency and high computational costs during network training. In future work, we will explore more strategies for reducing the model complexity and design a more lightweight network with real-time performance.

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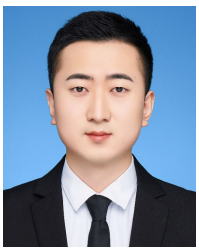
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